课程编号：10430494

课程时间：周三 第二大节 （ 9:50-11:25 ）六教6A316

周五 第一大节 （ 8:00- 9:35 ）六教6A316

大学物理B(2)（电磁学、光学、量子物理）

主讲：季帅华

邮箱：shji@mail.tsinghua.edu.cn

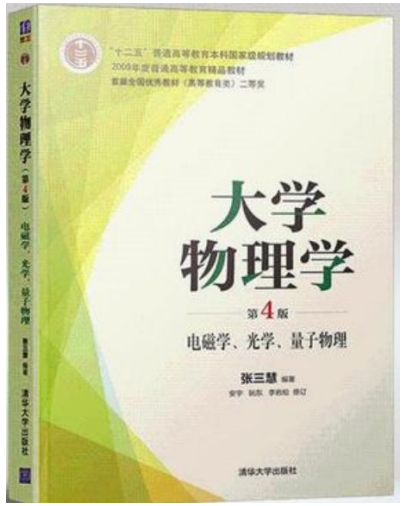
电话：62797539

办公室：物理系W341

助教：方业健

邮箱：fangyj25@mails.tsinghua.edu.cn

教材：



大学物理学（电磁学）
大学物理学（光学、量子物理）张三慧编著

参考书目：

《电磁学与电动力学》上册（胡友秋，程福臻，叶邦角编）

《电磁学》（贾起民，郑永令，陈暨耀编）

《新概念物理教程—电磁学》（赵凯华，陈熙谋编）

《光学》（崔宏滨，李永平，段开敏编）

《新概念物理教程—光学》（赵凯华，钟锡华编）

《原子物理学》（杨福家编）

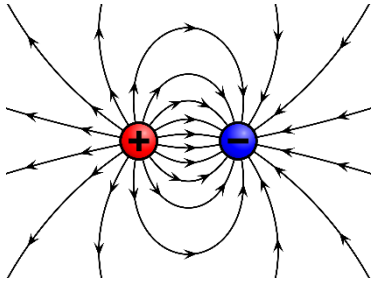
《费恩曼物理学讲义》第3卷（量子物理）（潘笃武，李洪芳译）

作业：周一交作业；作业：网络学堂上传

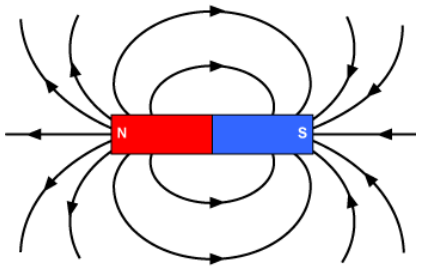
考试：期中第8周；期末第17/18周。

评分：考试约90%，作业约10%

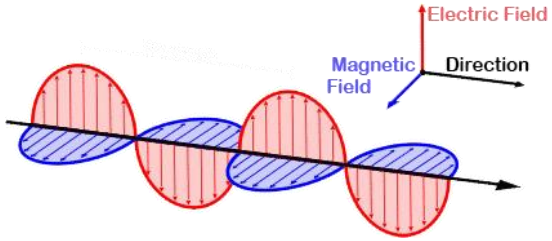
课程进程	
电磁学	静电场
	电势
	静电场中的导体
	静电场中的电介质
	稳恒电流
	磁场和它的源
	磁力
	磁场中的磁介质
	电磁感应
	麦克斯韦方程组和电磁波
期中考试	
光学	光的干涉
	光的衍射
	光的偏振
量子物理	波粒二象性
	薛定谔方程
	原子中的电子
	固体中的电子
	核与粒子物理
期末考试	



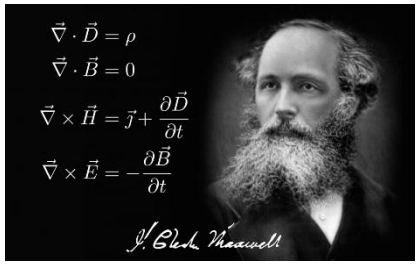
电场



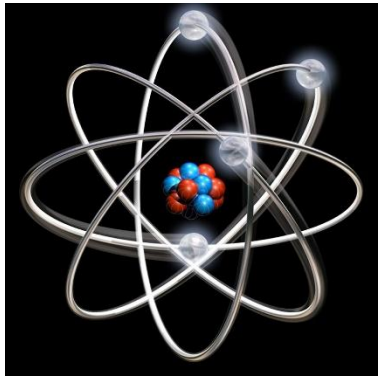
磁场



电磁波



麦克斯韦



量子力学

奠基者

投票(匿名) 最多可选3项

Setting

大学物理学习过程中，哪些你认为是重要的

A 数学方法和解题技巧

B 物理概念和物理图像

C 经典例题的讲解

D 建立实际问题的物理模型

E 学生之间以及师生之间的相互讨论

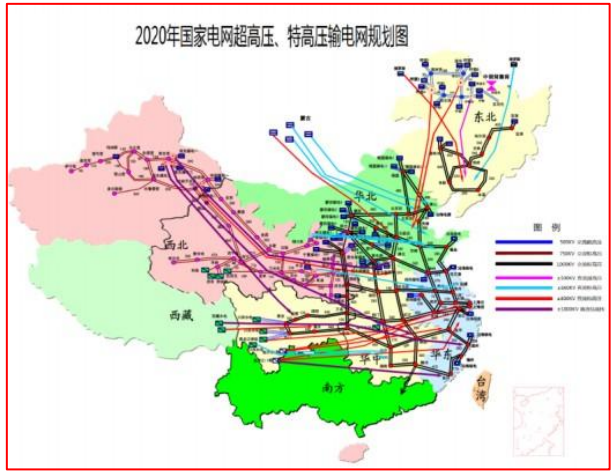
F 较难的课后习题

Submit

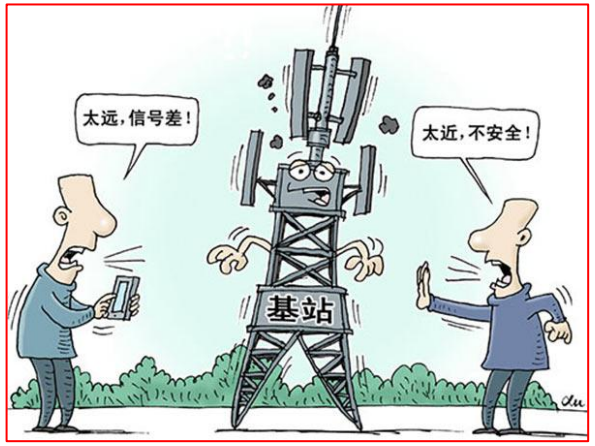
电 磁 学

特斯拉线圈

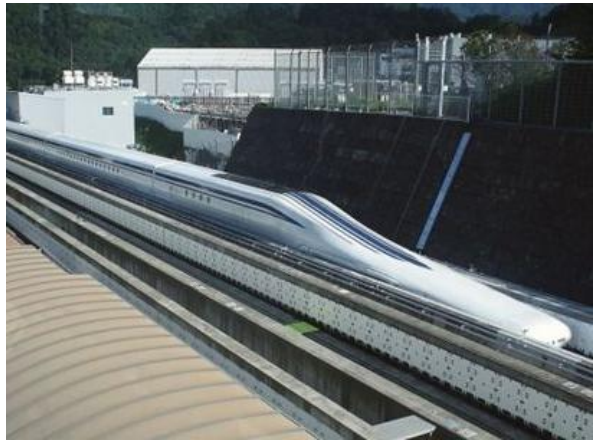
特高压国家电网



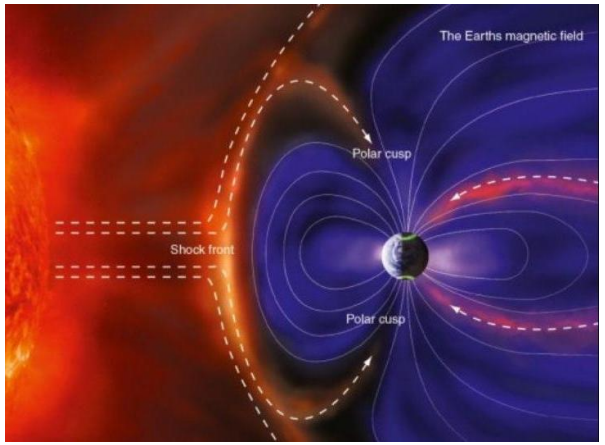
国家电网规划



手机基站



日本磁悬浮列车
（603公里/小时）



太阳风

电磁学

一. 静电场

二. 电势

三. 静电场中的导体

四. 静电场中的电介质

五. 稳恒电流

六. 磁场和它的源

七. 磁力

八. 磁场中的磁介质

九. 电磁感应

十. 麦克斯韦方程组和电磁波

电磁学与力、热学区别

力学：粒子的运动 (单个或多个)

$$\vec{F} = \frac{d\vec{p}}{dt} \qquad \vec{p},\ \frac{1}{2}mv^2,\ \vec{r}\times\vec{p},\ \ldots\ldots$$

热学: 大量粒子的无规运动规律

统计方法 和 宏观热力学

电磁学: 带电粒子? 只是场量的源而已

电磁学：场量的性质

$$\vec{E}(x,y,z,t), \quad \vec{B}(x,y,z,t)$$

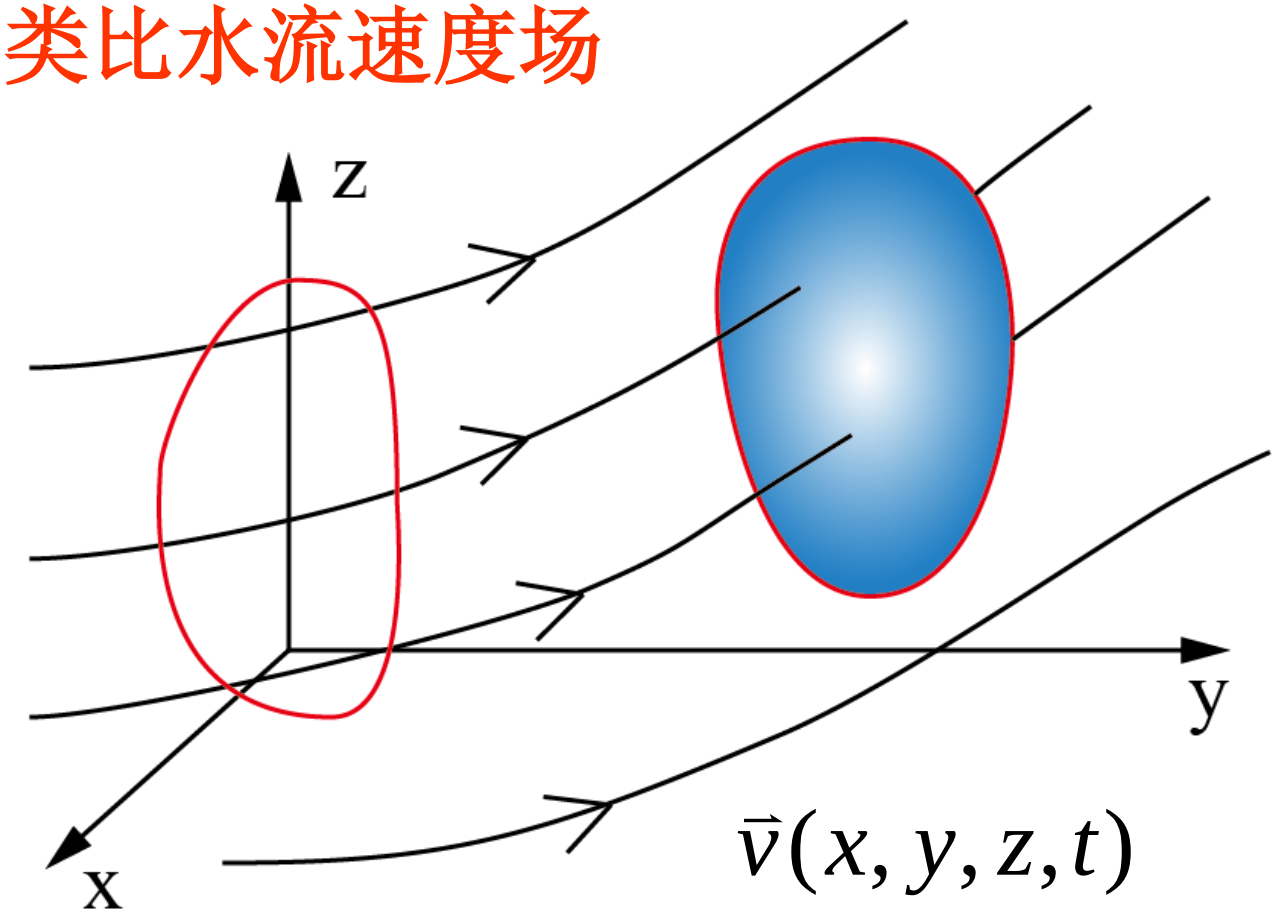
法拉第最先提出来的概念

场：现代物理学的核心概念

如何描述电场和磁场的性质

$$\vec{E}(x,y,z,t),\quad \vec{B}(x,y,z,t)$$

类比水流速度场

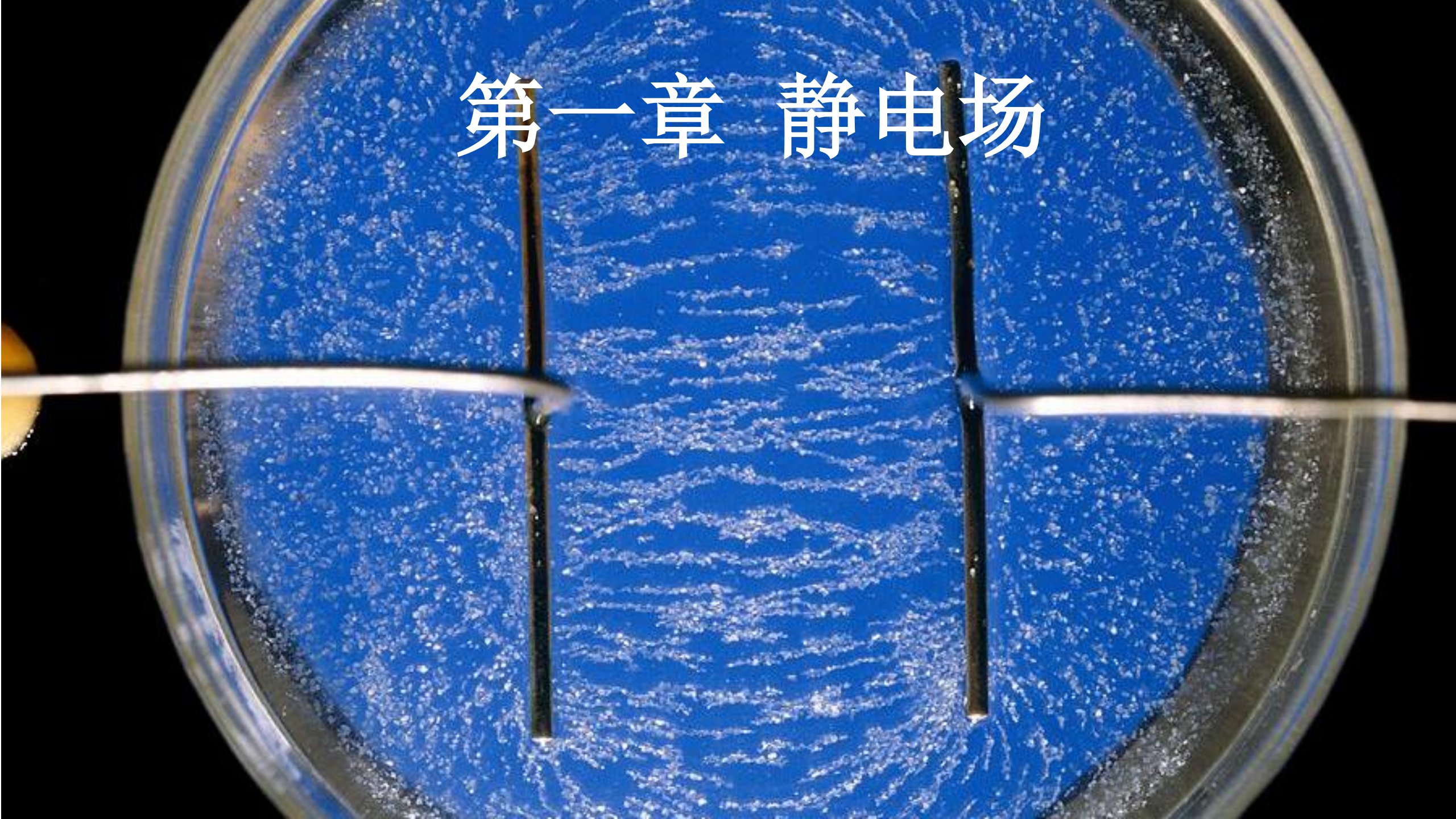


数学描述矢量场的方法

闭合曲面的通量

闭合路径的环量

第一章 静电场



- 核心概念
  - 电场
  - 电通量
- 核心问题
  - 点电荷的电场是什么样的？
  - 什么是电场的叠加原理？
  - 什么是电通量？
  - 什么是高斯定律？如何应用高斯定律？

第一章 静电场

1.1 电荷

1.2 库仑定律

1.3 电场 电场强度

1.4 点电荷电场及叠加原理

1.5 电通量

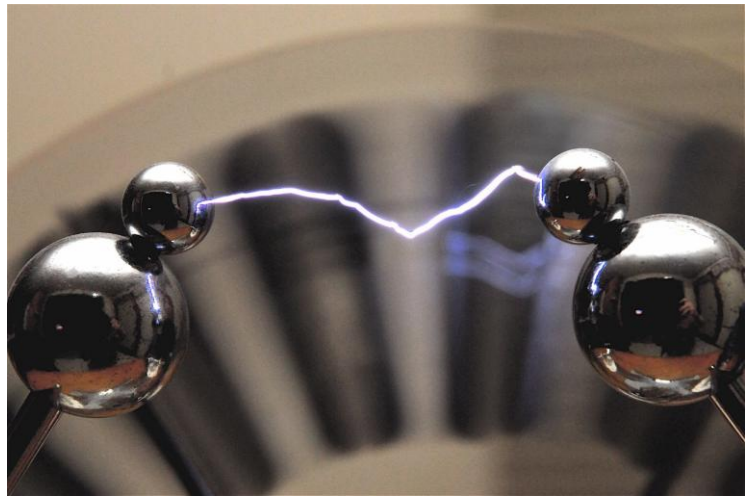
1.6 静电场的高斯定律证明

1.7 高斯定律和电场线

1.8 高斯定理的应用

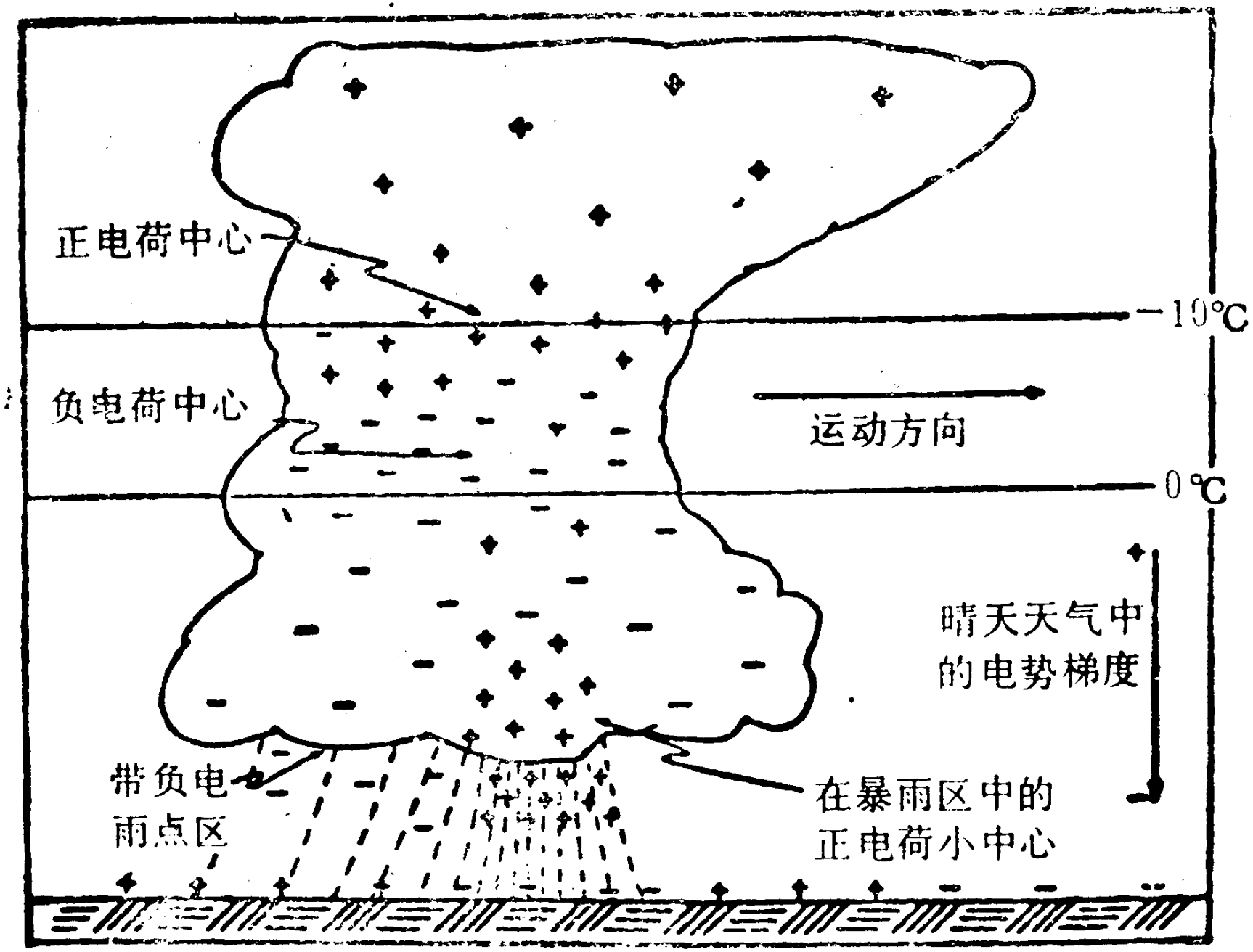
1.1　电荷

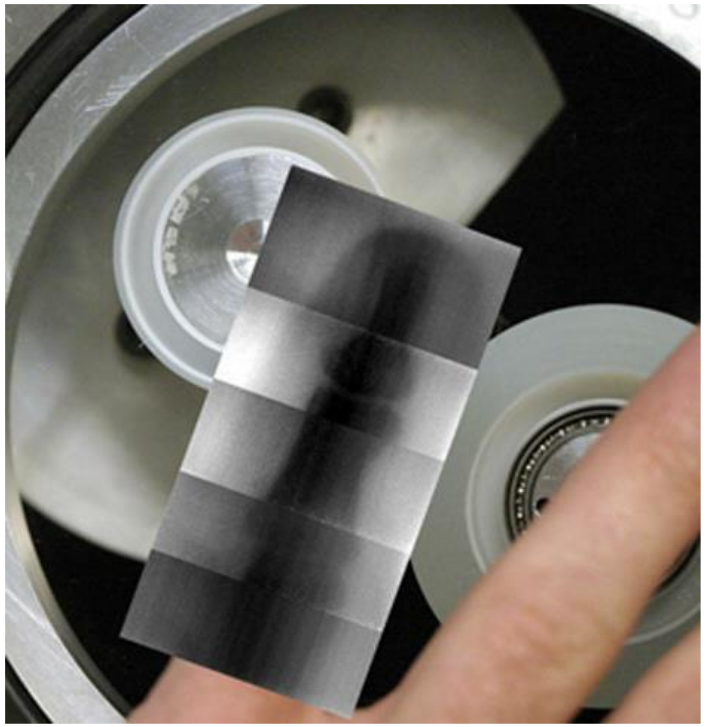
❖　两种（正电荷负电荷）



“天上的闪电”和电火花是同一物理现象

富兰克林


正电荷中心
负电荷中心
-10℃
0℃
运动方向
晴天天气中
的电势梯度
带负电
雨点区
在暴雨区中的
正电荷小中心



Nature 455, 1089 (2008)

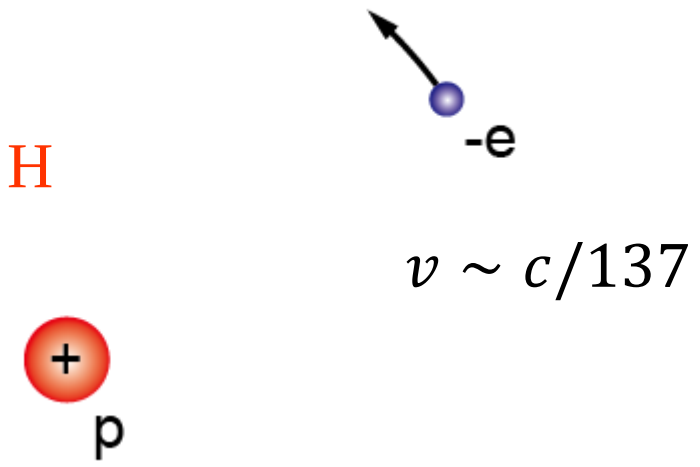
水滴皇冠

❖ 电荷量子化（charge quantization ）

1906-1917年，密立根用液滴法首先从实验上证明了，微小粒子带电量的变化不连续。

$$\boxed{Q = Ne}$$

❖ 电量的相对论不变性



$v \sim c/137$

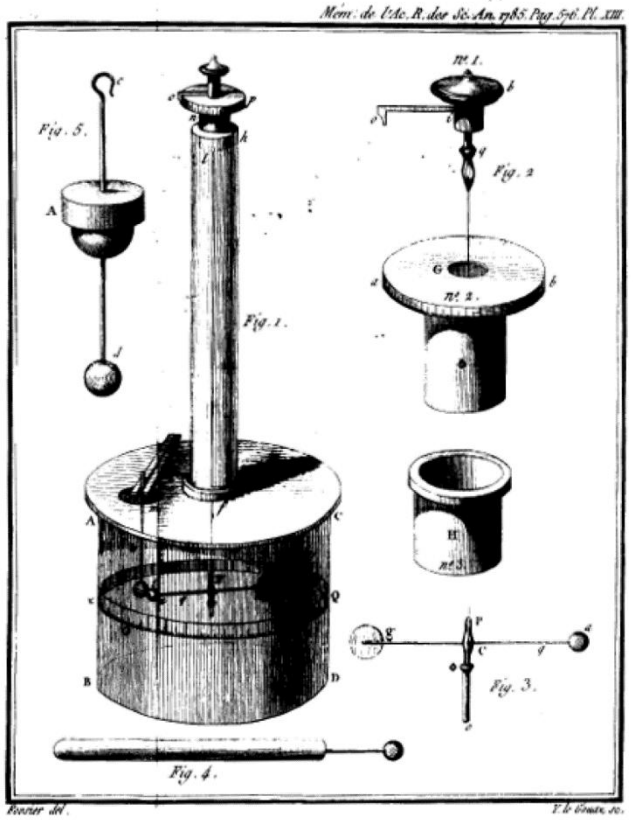
❖ 电荷守恒定律

电荷守恒定律是物理学中**普遍的**基本定律

$$\boxed{\sum Q_i = c}$$

局域守恒

1.2　库仑定律

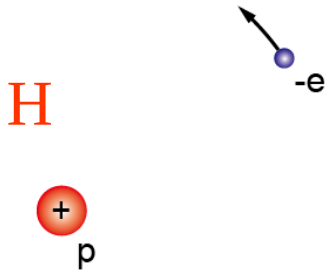


库仑定律：在真空中，　两个静止**点电荷**之间的相互作用力大小，与它们的电量的乘积成正比，与它们之间距离的平方成反比；作用力的方向沿着它们的联线，同号电荷相斥，异号电荷相吸。

$$\boxed{\vec{f} = \frac{q_1 q_2}{4\pi\varepsilon_0 r^2}\hat{r}}$$

1785年，库仑通过扭称实验得到。

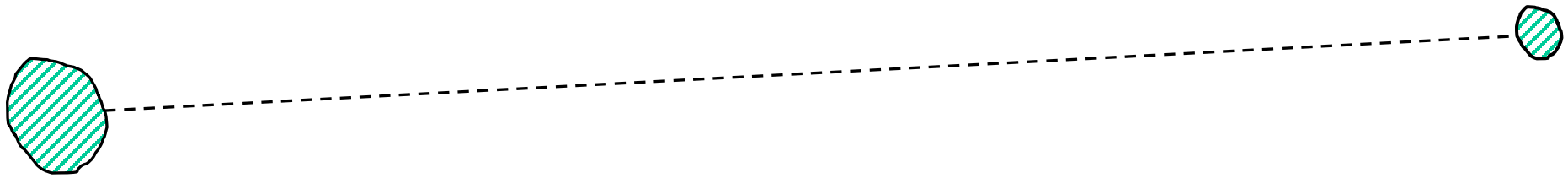
(1) 库仑力很强



$$\frac{f_{电力}}{f_{万有引力}} \sim 10^{39}$$

(2) 基本实验规律

(3) 点电荷　　理想模型

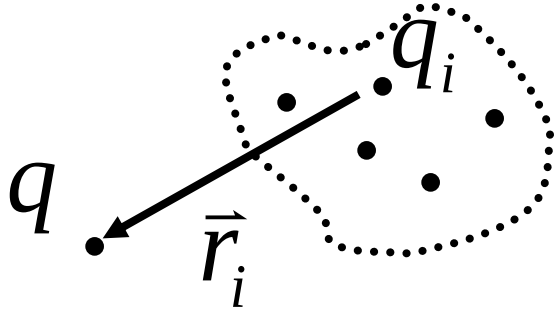


(4) 适用范围：宏观 – 微观

在原子核内($<10^{-15}$m)

更小范围内倾向于反平方律仍然成立

(5) 电力叠加原理



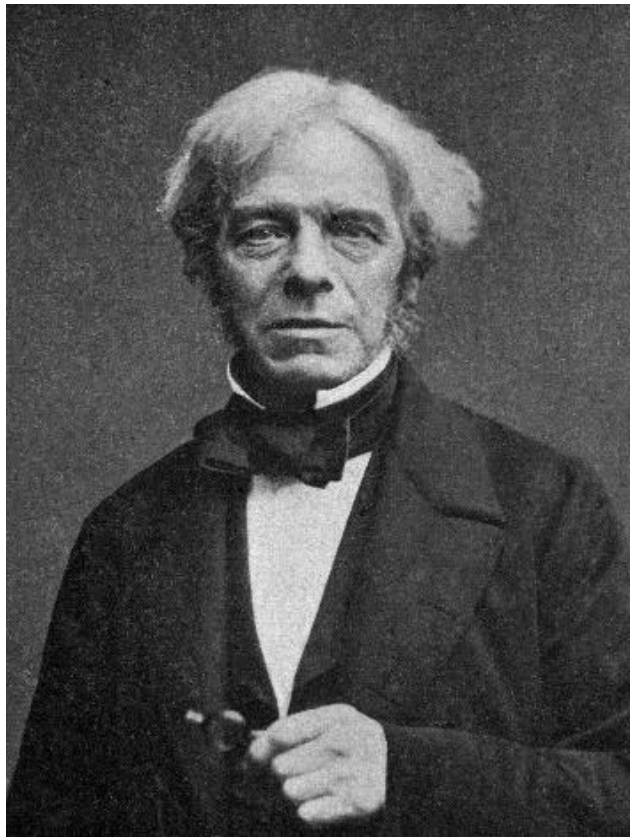
$$\boxed{\vec{f} = \sum_i \vec{f}_i}$$

1.3 电场 电场强度

早期：超距作用

后来：法拉第提出近距作用

并提出力线和场的概念

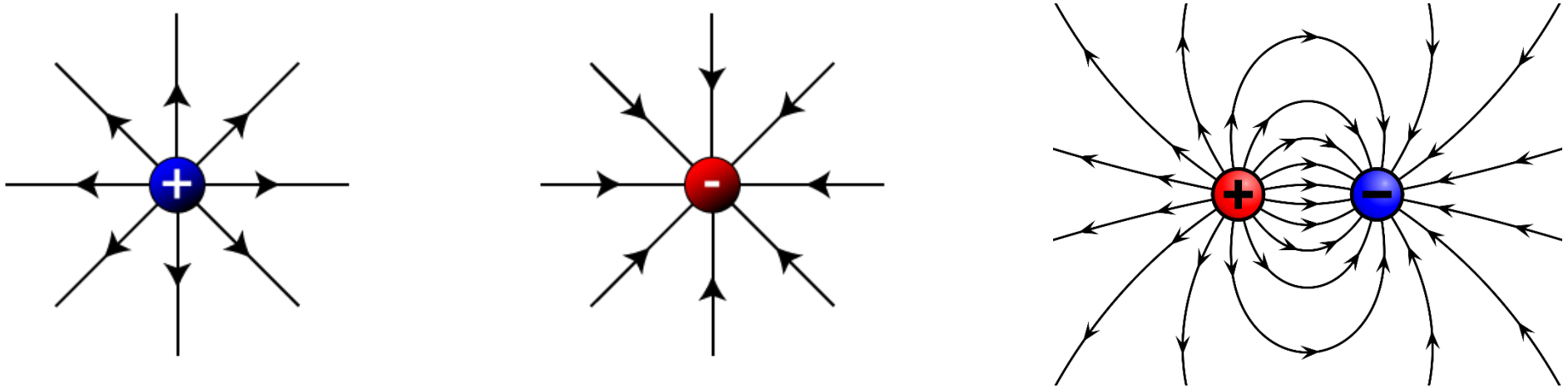


Michael Faraday
(1791-1867)

一. 电场

- 电荷在其周围产生电场。
- 电荷在电场中受力

静电场: 静止的电荷产生的电场



二. 电场强度 (electric field strength)

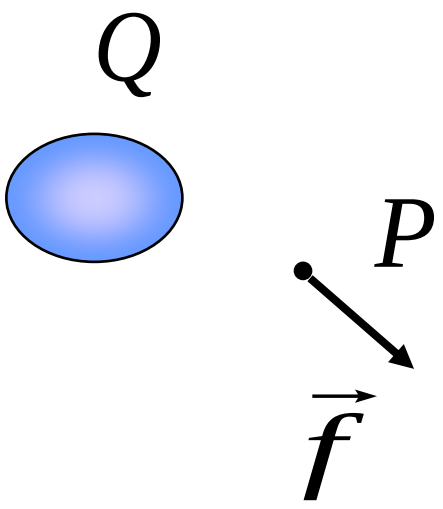
描述场中各点电场的强弱的物理量

空间带电体　电量为 Q

试验电荷

电量充分地小

线度足够地小



试验表明：确定场点　比值 $\vec{f}/q$　与试验电荷无关

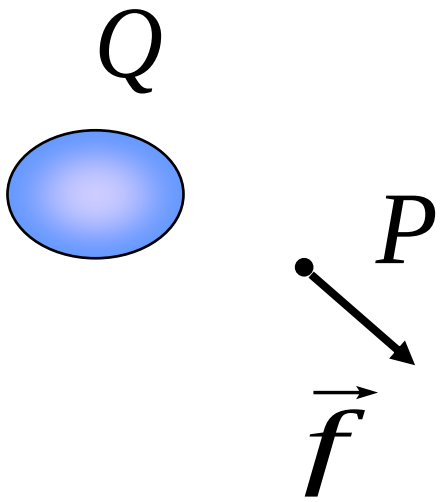
电场强度定义

$$\boxed{\vec{E} = \frac{\vec{f}}{q}}$$

$$\boxed{\vec{E} = \vec{E}(x,\ y,\ z)}$$

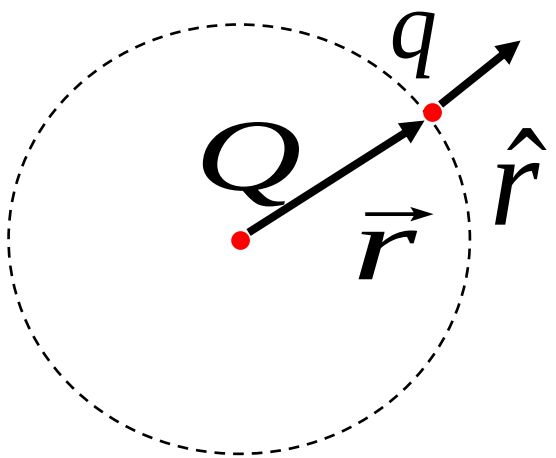
❖ 点电荷在电场中受力

$$\boxed{\vec{f} = q\vec{E}}$$



1.4　点电荷电场及叠加原理

1. 点电荷的场强公式



$$\boxed{\vec{f} = \frac{Qq}{4\pi\varepsilon_0 r^2}\hat{r}}$$

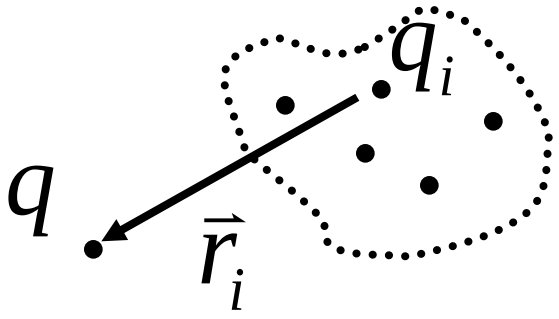
$$\boxed{\vec{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\hat{r}}$$

$$\boxed{\vec{E} = \frac{\vec{f}}{q}}$$

球对称

2. 场强叠加原理

带电体由 n 个点电荷组成



由电力叠加原理 $$\boxed{\vec{f} = \sum_{i=1}^{i=n} \vec{f}_i}$$

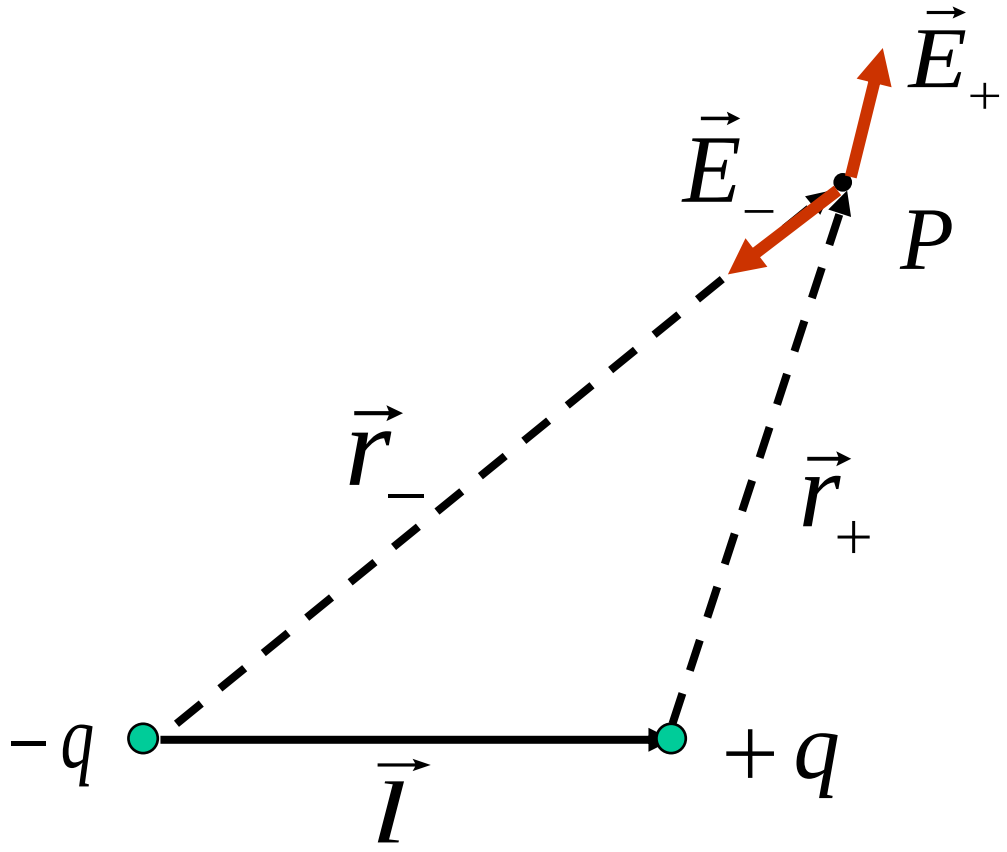
由场强定义 $$\vec{E} = \frac{\vec{f}}{q} = \frac{\sum_{i=1}^{n} \vec{f}_i}{q} = \sum_{i=1}^{n} \frac{\vec{f}_i}{q}$$

$$\boxed{\vec{E} = \sum_i \vec{E}_i}$$ 或 $$\boxed{\vec{E} = \sum_{i=1}^{n} \frac{q_i}{4\pi\varepsilon_0 r_i^2} \hat{r}_i}$$

3 电荷离散分布的点电荷集

例　电偶极子的场

一对等量异号电荷, 相距 l



点电荷场叠加

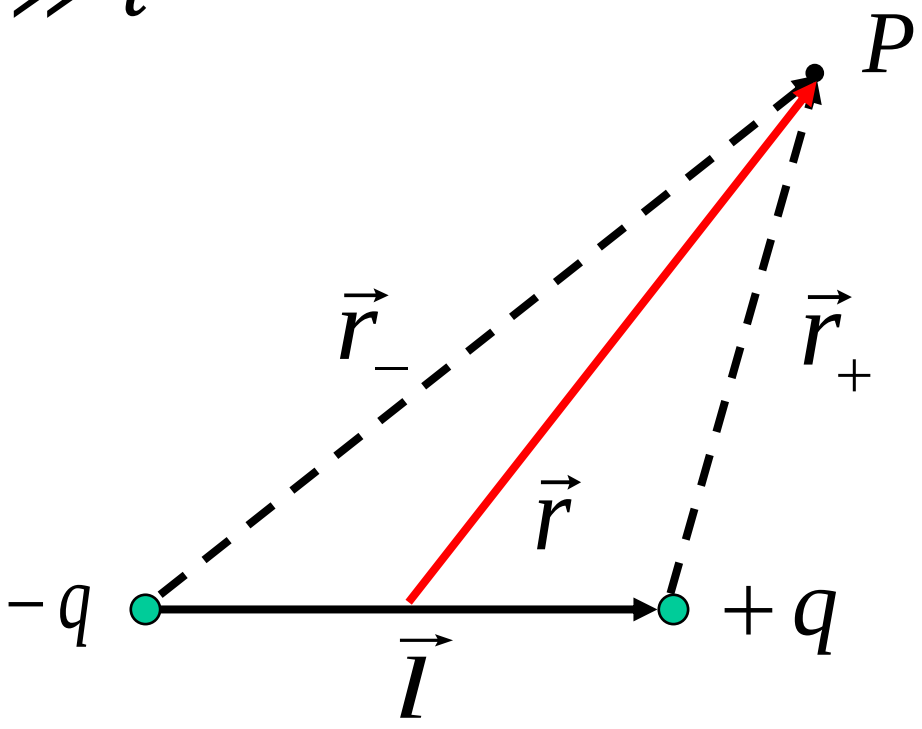
$$\vec{E} = \vec{E}_+ + \vec{E}_-$$

$$= \frac{q_+}{4\pi\,\varepsilon_0 r_+^2}\hat{r}_+ + \frac{q_-}{4\pi\,\varepsilon_0 r_-^2}\hat{r}_-$$

$$= \frac{q}{4\pi\varepsilon_0 r_+^2}\hat{r}_+ + \frac{-q}{4\pi\varepsilon_0 r_-^2}\hat{r}_-$$

$$= \frac{q}{4\pi\varepsilon_0}\left(\frac{\vec{r}_+}{r_+^3} - \frac{\vec{r}_-}{r_-^3}\right)$$

$r \gg l$

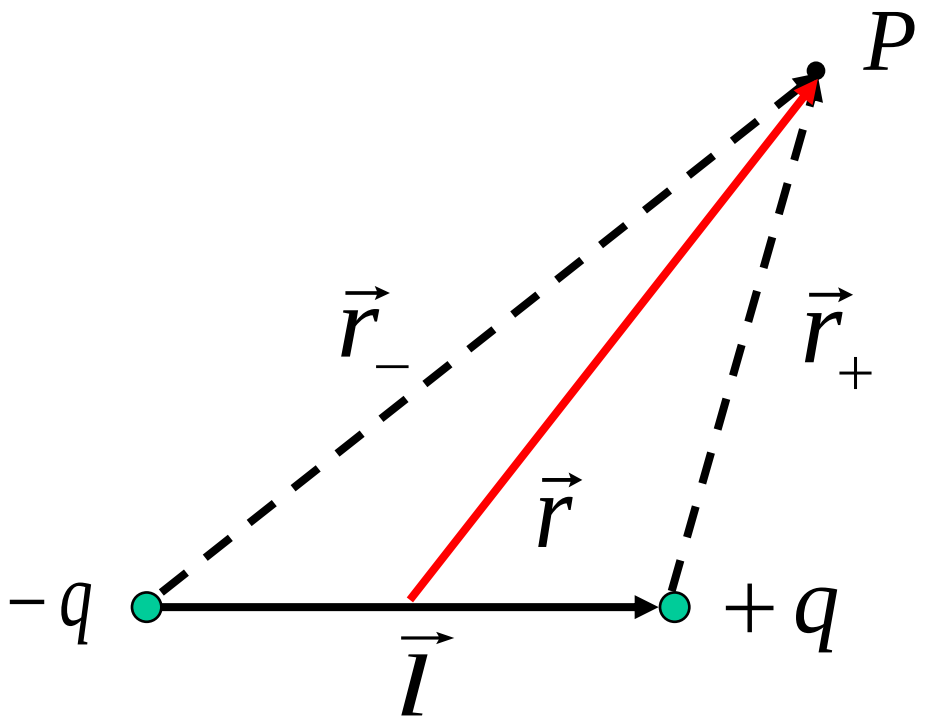


电偶极子 (electric dipole)

电偶极矩 (electric moment)

$$\boxed{\vec{p} = q\vec{l}}$$

$$\vec{E} = \frac{q}{4\pi\varepsilon_0}\left(\frac{\vec{r}_+}{r_+^3} - \frac{\vec{r}_-}{r_-^3}\right)$$



$$\vec{r}_+ = \vec{r} - \frac{\vec{l}}{2} \qquad \vec{r}_- = \vec{r} + \frac{\vec{l}}{2}$$

$$r_+^2 = r^2 + \frac{l^2}{4} - \vec{r}\cdot\vec{l} \qquad r_-^2 = r^2 + \frac{l^2}{4} + \vec{r}\cdot\vec{l}$$

$$r_+^{-3} = r^{-3}\left[1 + \frac{l^2}{4r^2} - \frac{\vec{r}\cdot\vec{l}}{r^2}\right]^{-3/2}$$

$$(1+x)^\alpha \approx 1 + \alpha x + \frac{1}{2}\alpha(\alpha-1)x^2 + \ldots\ldots$$

$$r \gg l$$

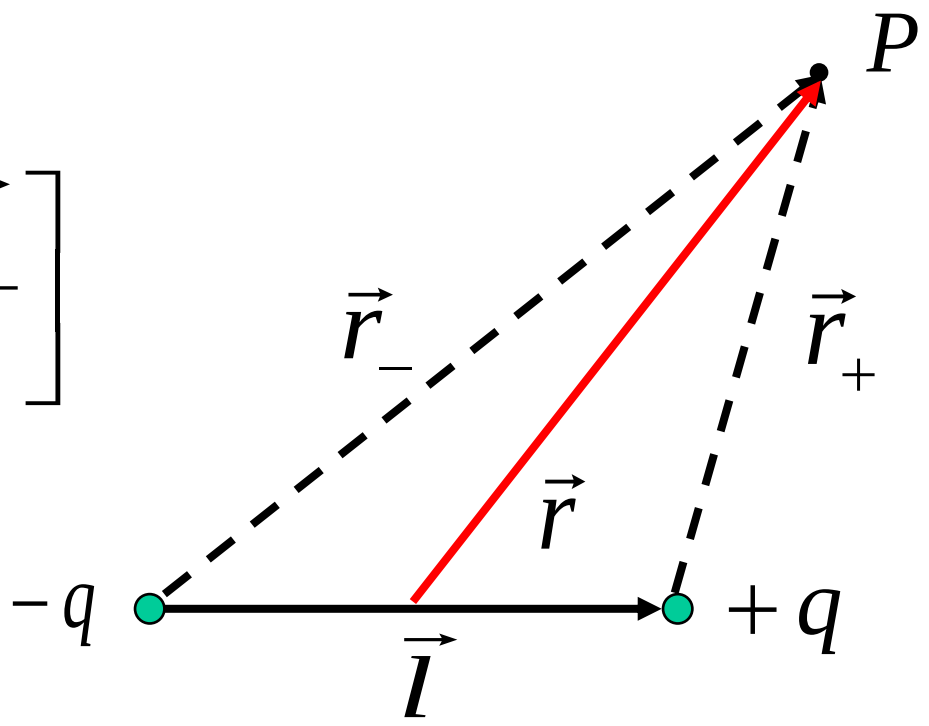
$$r_+^{-3} = r^{-3}\left(1 + \frac{3}{2}\frac{\vec{r}\cdot\vec{l}}{r^2}\right) \qquad r_-^{-3} = r^{-3}\left(1 - \frac{3}{2}\frac{\vec{r}\cdot\vec{l}}{r^2}\right)$$

$$\vec{E}=\frac{q}{4\pi\varepsilon_0}\left(\frac{\vec{r}_+}{r_+^3}-\frac{\vec{r}_-}{r_-^3}\right)$$

$$\vec{E}=\frac{q}{4\pi\varepsilon_0 r^3}\left[\vec{r}_+-\vec{r}_-+\left(\vec{r}_++\vec{r}_-\right)\frac{3}{2}\frac{\vec{r}\cdot\vec{l}}{r^2}\right]$$

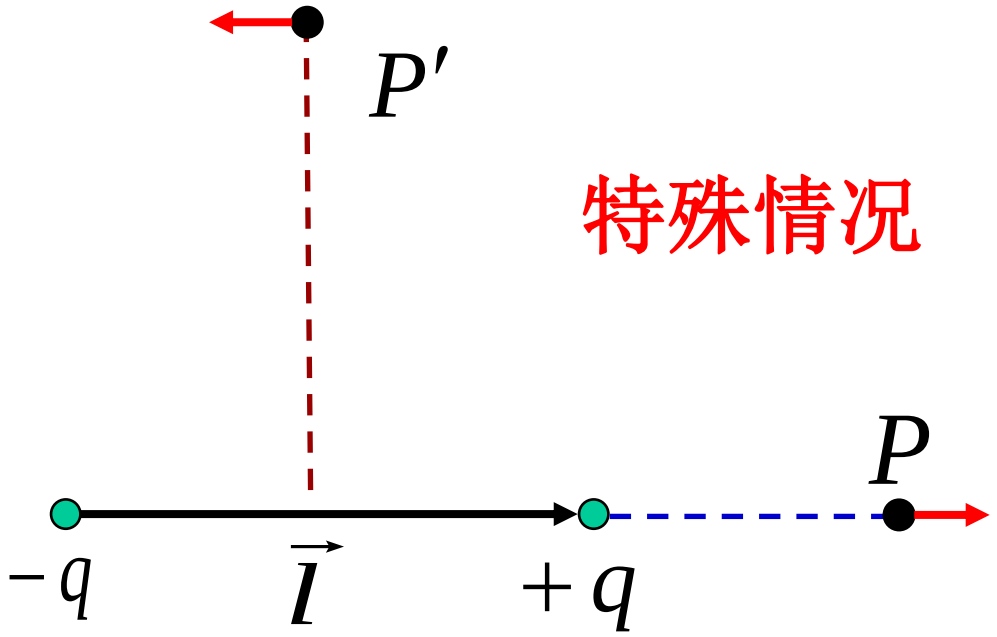
$$\vec{r}_+-\vec{r}_-=-\vec{l}$$

$$\vec{r}_++\vec{r}_-=2\vec{r}$$



电偶极子电场

$$\boxed{\vec{E}=\frac{1}{4\pi\varepsilon_0 r^3}\left[-\vec{p}+3\left(\hat{r}\cdot\vec{p}\right)\hat{r}\right]}$$



特殊情况

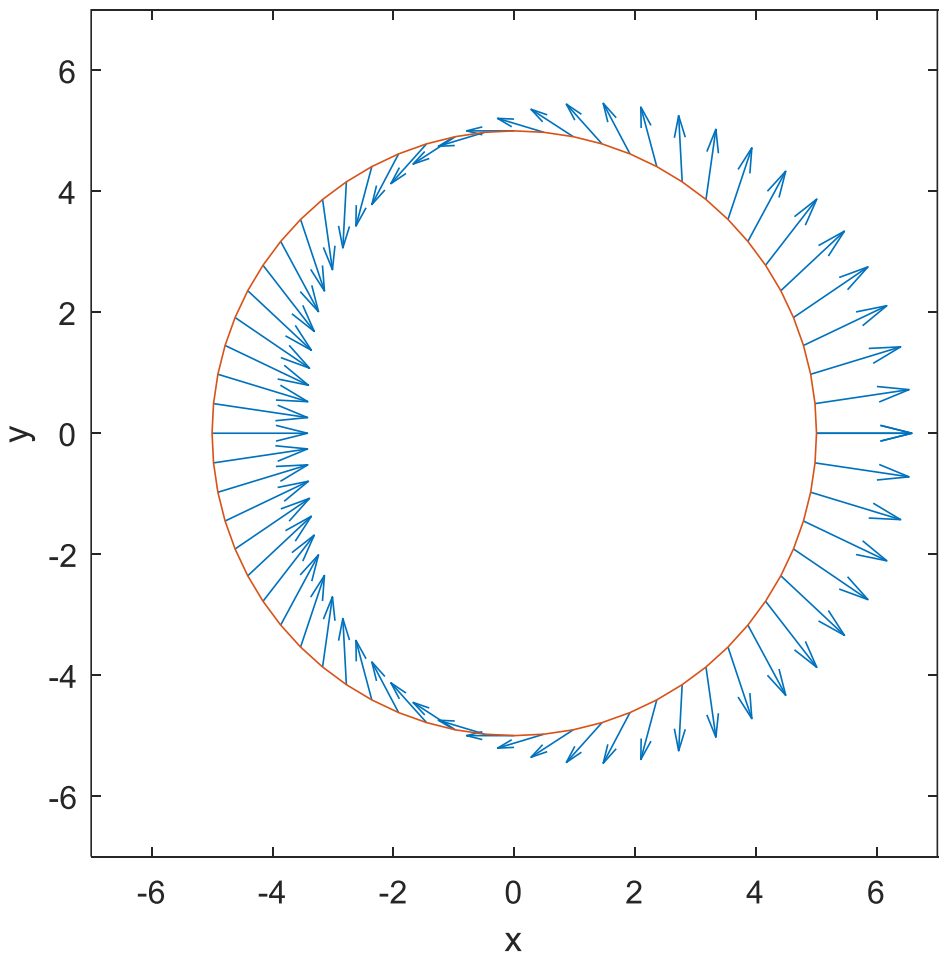
❖ 连线上

$$\vec{E} = \frac{2\vec{p}}{4\pi\varepsilon_0 r^3}$$

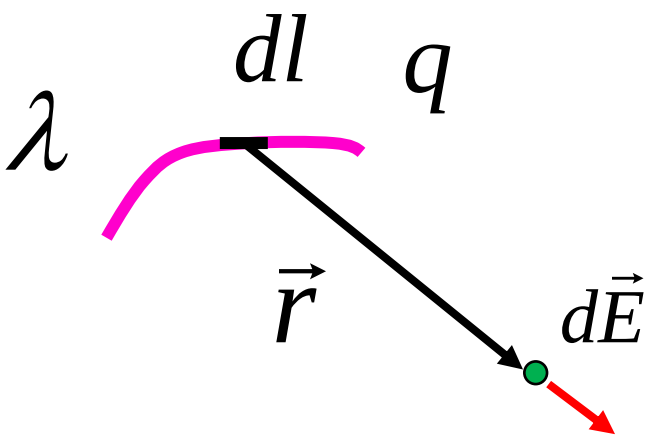
❖ 垂直连线上

$$\vec{E} = \frac{-\vec{p}}{4\pi\varepsilon_0 r^3}$$

电偶极子的电场



4 电荷连续分布的带电线



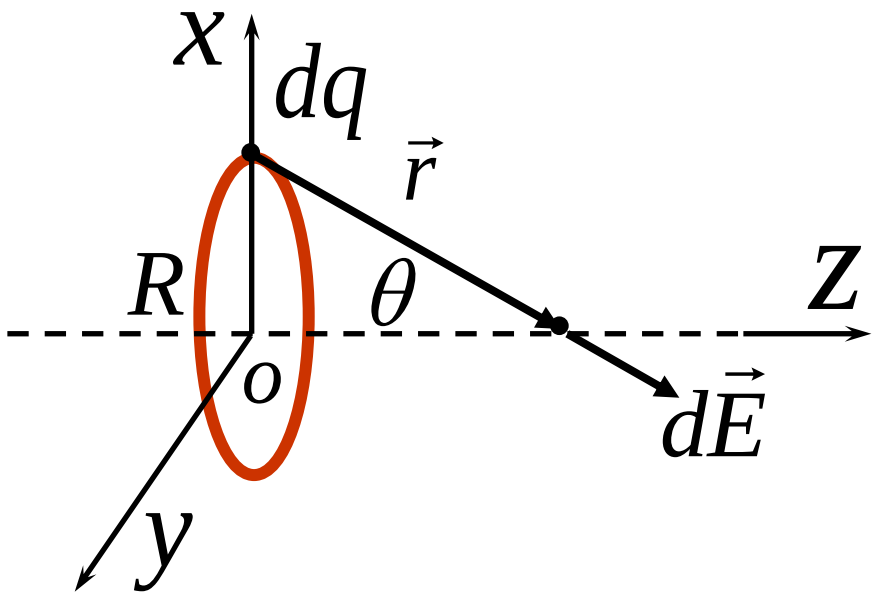
线电荷密度

$$\boxed{\lambda = \frac{dq}{dl}}$$

场强叠加原理

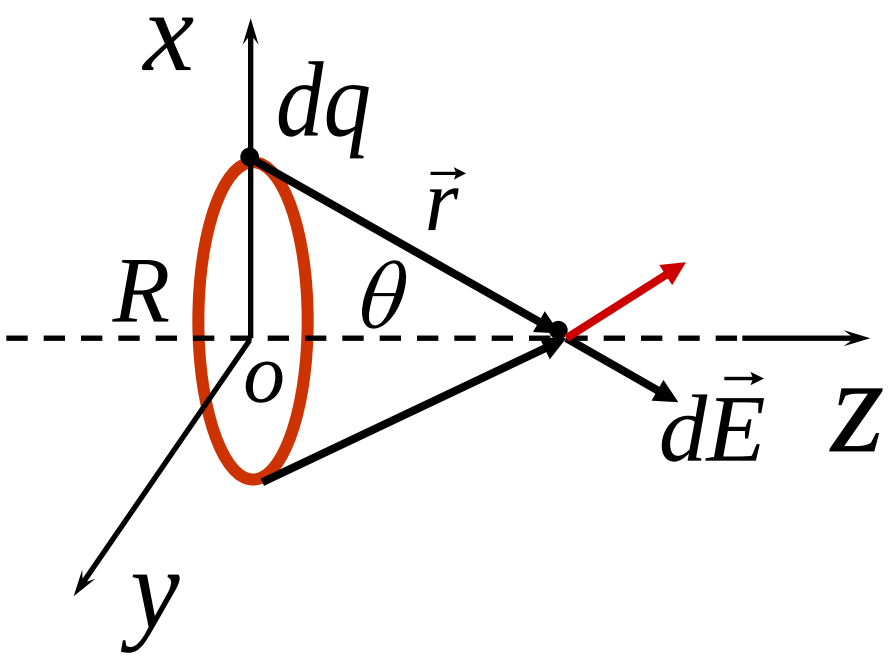
$$\boxed{\vec{E} = \int_{(q)} \frac{1}{4\pi\varepsilon_0} \frac{\lambda dl}{r^2} \hat{r}}$$

例 均匀带电圆环轴线上的场



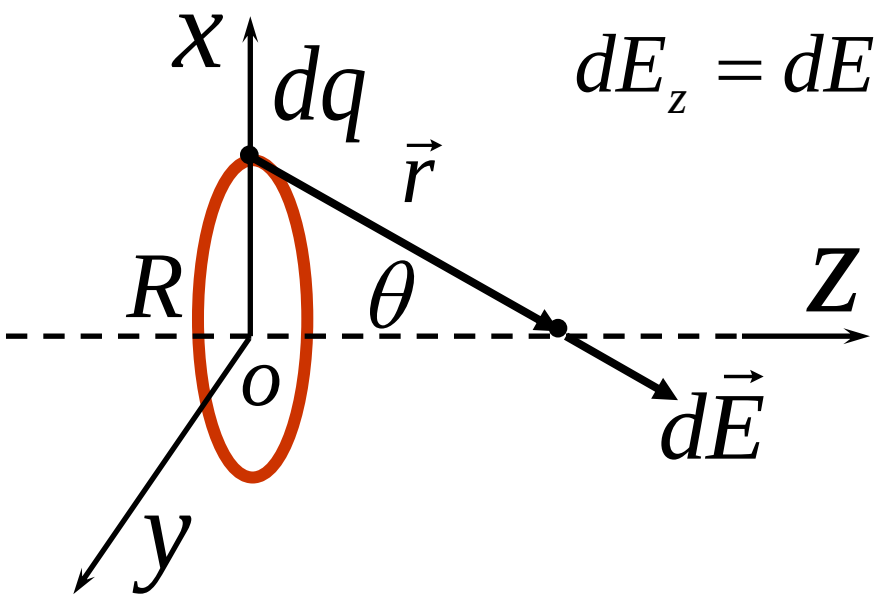
解：在圆环上任取电荷元 $dq = \lambda dl$

$$\boxed{d\vec{E} = \frac{\lambda dl}{4\pi\varepsilon_0 r^2}\hat{r}}$$



由对称性分析知, 垂直z 轴的场强为 0

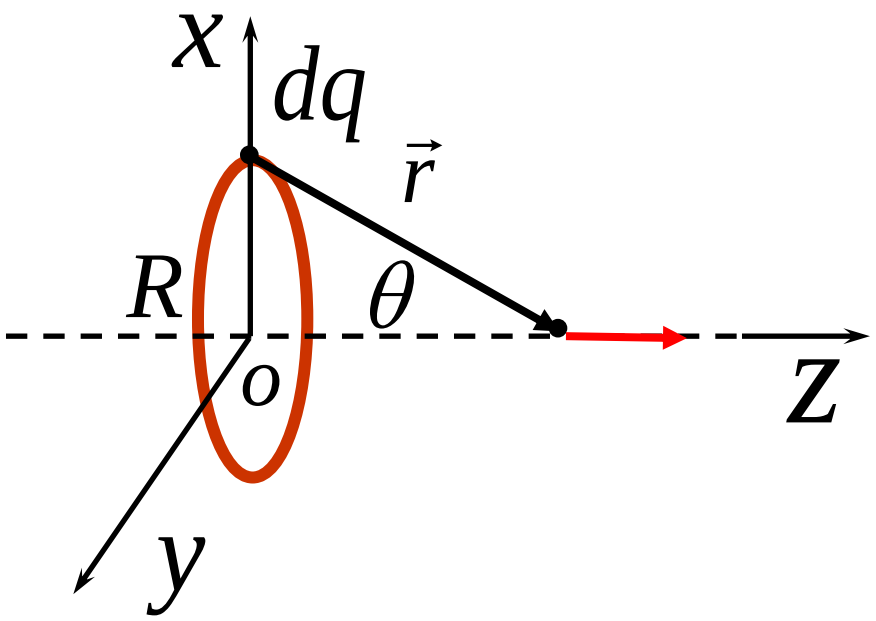
$$\boxed{\vec{E} = E_z \hat{z}}$$



$$dE_z = dE\cos\theta$$

$$E = E_z = \int_{(q)} \frac{\lambda dl}{4\pi\varepsilon_0 r^2}\cos\theta$$

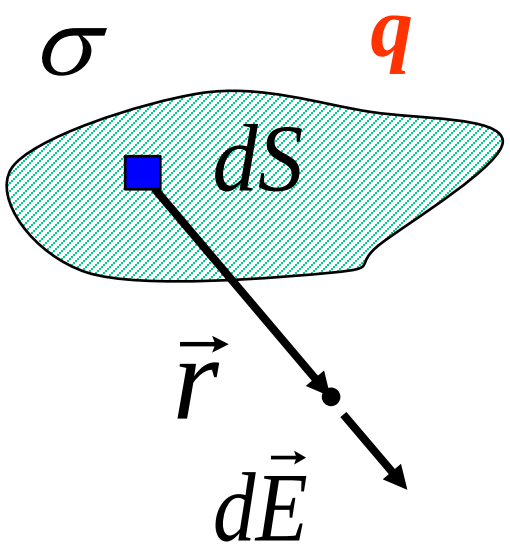
$$= \frac{\lambda\cos\theta}{4\pi\varepsilon_0 r^2}\int_{(q)} dl$$



$$E=\frac{zQ}{4\pi\varepsilon_0\left(z^2+R^2\right)^{3/2}}$$

若 $z \gg R$ $\quad E=\dfrac{Q}{4\pi\varepsilon_0 z^2}$ 点电荷

5 电荷连续分布的带电面



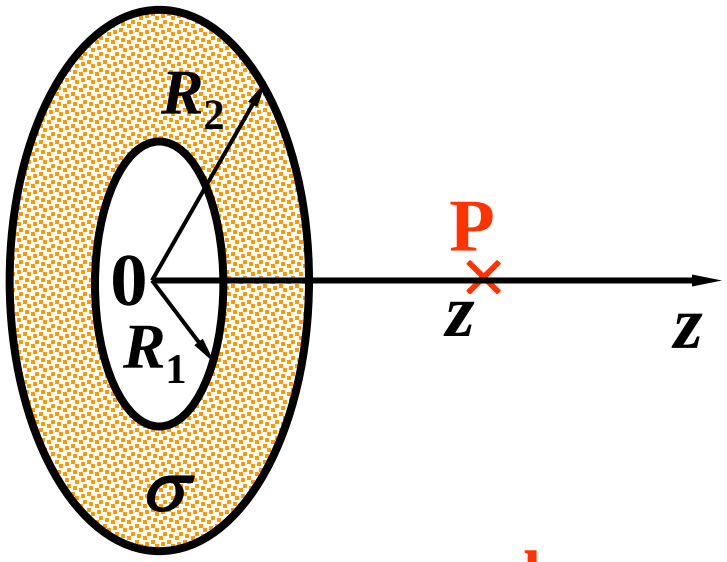
面电荷密度

$$\boxed{\sigma = \frac{dq}{dS}}$$

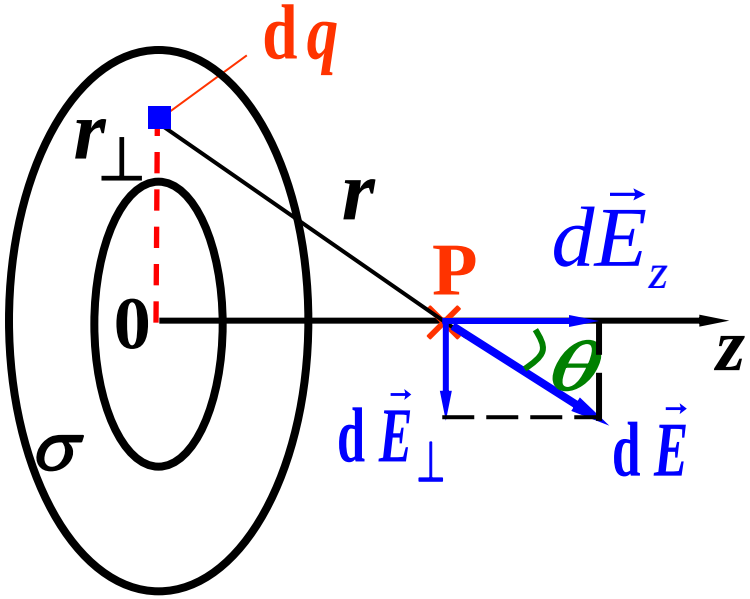
场强叠加原理

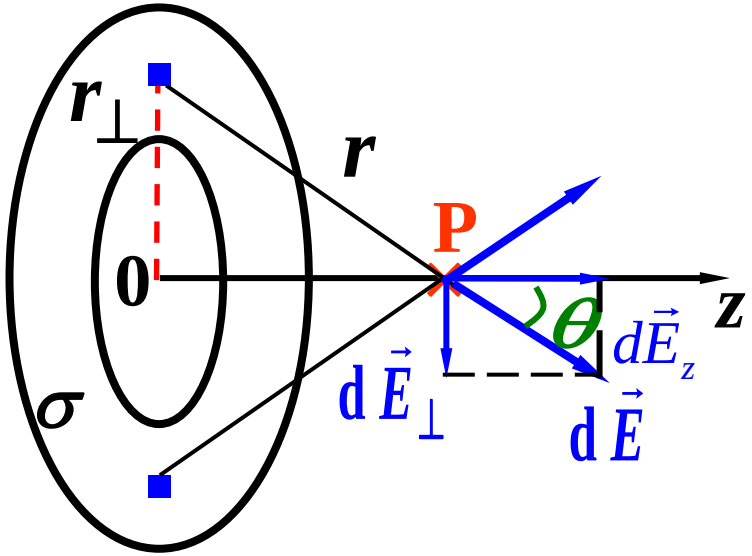
$$\boxed{\vec{E} = \int_{(q)} \frac{1}{4\pi\varepsilon_0} \frac{\sigma dS}{r^2} \hat{r}}$$

例 均匀带电环面中轴线上场强



解：



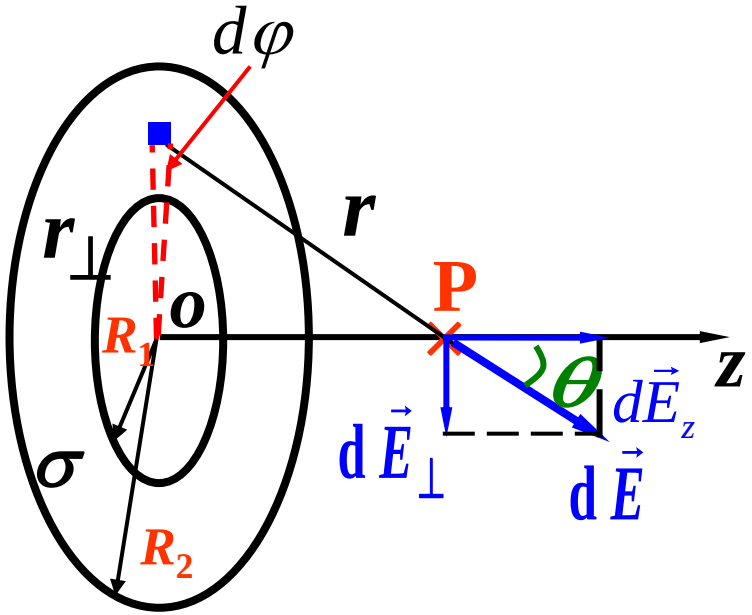


$$\boxed{\vec{E} = E_z \hat{z}}$$

由对称性可知电场方向沿z轴方向

$$\vec{E} = \int d\vec{E}_z + \int d\vec{E}_\perp$$

$$dE_z = \frac{dq}{4\pi\varepsilon_0 r^2}\cos\theta = \frac{zdq}{4\pi\varepsilon_0 r^3}$$



$$E_z = \int_0^{2\pi}\int_{R_1}^{R_2} \frac{z\sigma r_\perp d\varphi dr_\perp}{4\pi\varepsilon_0\left(z^2 + r_\perp^2\right)^{3/2}}$$

$$= \frac{z\sigma}{2\varepsilon_0}\left(\frac{1}{\sqrt{z^2 + R_1^2}} - \frac{1}{\sqrt{z^2 + R_2^2}}\right)$$

讨论：

$$E_z = \frac{\sigma z}{2\varepsilon_0}\left[\frac{1}{\sqrt{z^2+R_1^2}} - \frac{1}{\sqrt{z^2+R_2^2}}\right]$$

① 量纲正确；

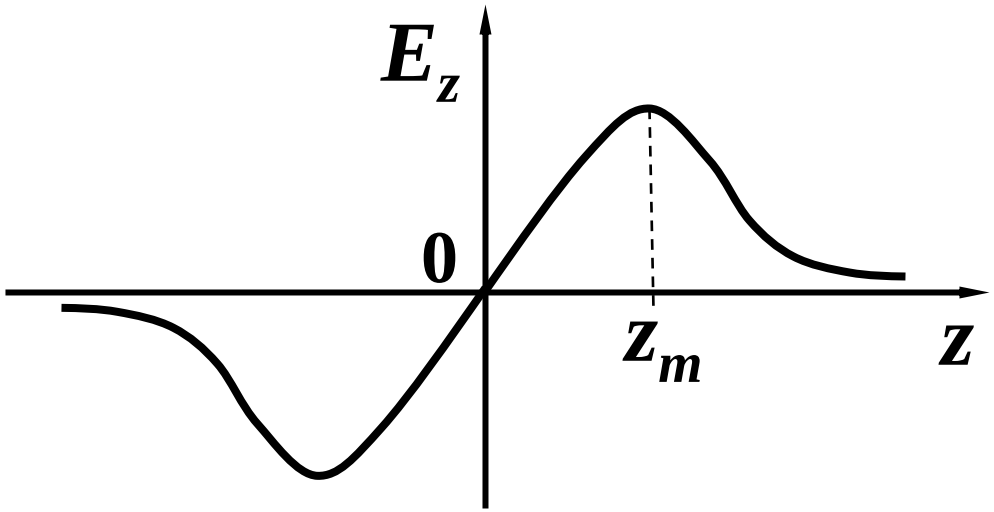
② 令 $z >> R_2$，

$$\frac{1}{\sqrt{z^2+R^2}} = \frac{1}{z\sqrt{1+R^2/z^2}} = \frac{1}{z}\left(1+\frac{R^2}{z^2}\right)^{-\frac{1}{2}} \approx \frac{1}{z}\left(1-\frac{R^2}{2z^2}\right)$$

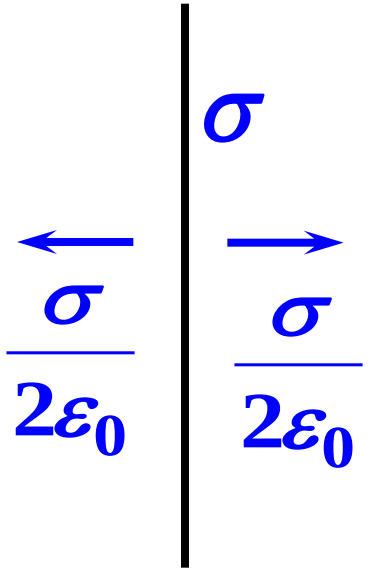
$$E_z \approx \frac{\sigma}{2\varepsilon_0}\cdot\frac{R_2^2-R_1^2}{2z^2} = \underline{\frac{q}{4\pi\varepsilon_0 z^2}}\ ,$$ 合理。

点电荷电场

③ E的分布：

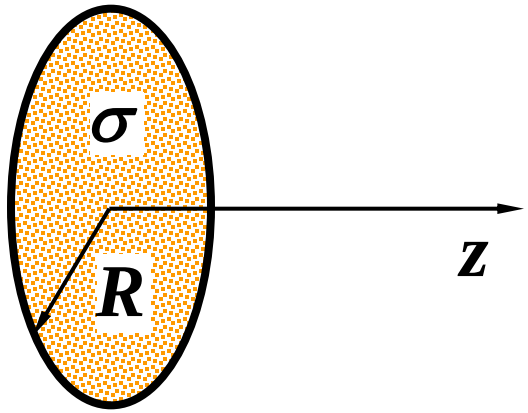


④ $R_1 \to 0$， $R_2 \to \infty$，此为均匀带电无限大平面：



$$\boxed{E = |E_x| = \frac{\sigma}{2\varepsilon_0}}$$

⑤ $R_1 \to 0$， $R_2 = R$ ，此为均匀带电圆盘情形：

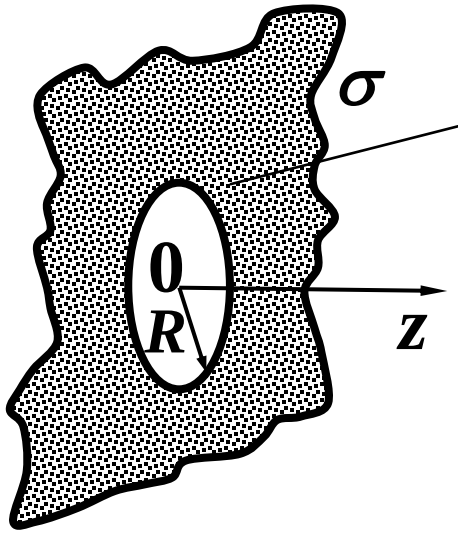


$$E_z = \frac{\sigma z}{2\varepsilon_0}\left[\frac{1}{|z|} - \frac{1}{\sqrt{z^2 + R^2}}\right]$$

$$= \frac{\sigma}{2\varepsilon_0} \cdot \frac{z}{|z|}\left(1 - \frac{|z|}{\sqrt{z^2 + R^2}}\right)$$

⑥ 思考

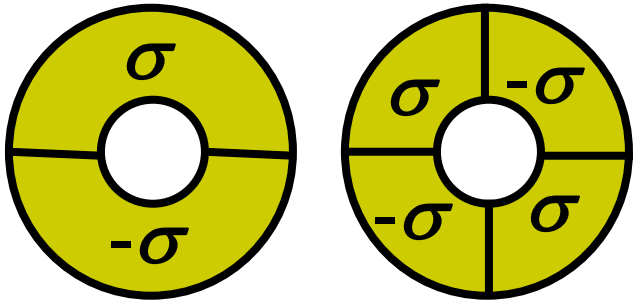
（a）z轴上E =?



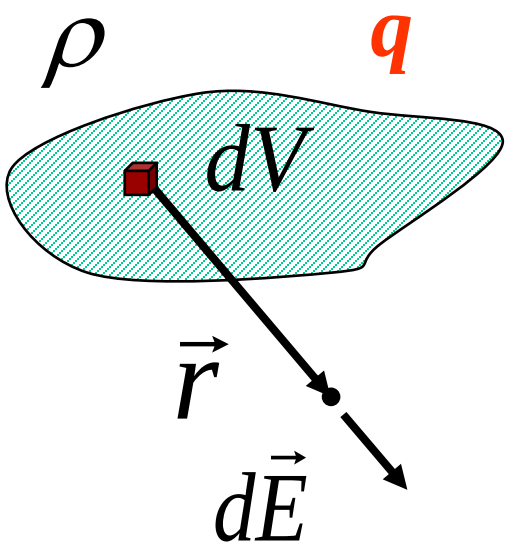
挖一圆孔的无限大均匀带电平面

（b）z >>电荷线度处，

E 有何特点？



6 电荷连续分布的带电体



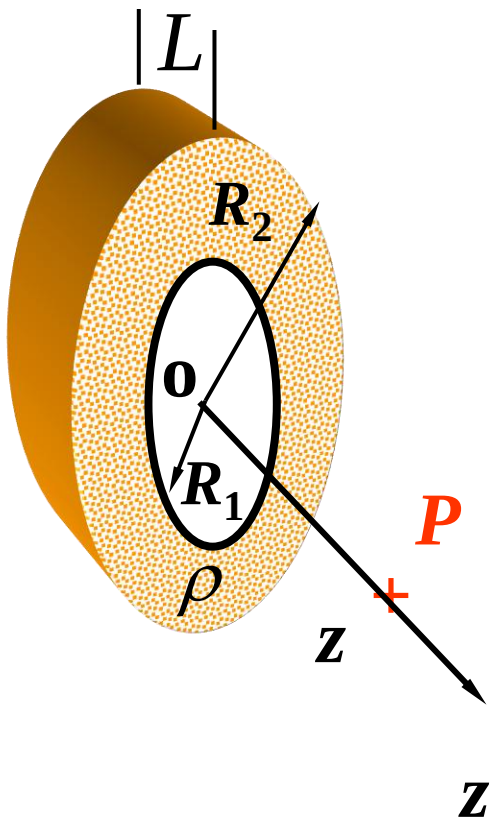
体电荷密度

$$\boxed{\rho = \frac{dq}{dV}}$$

场强叠加原理

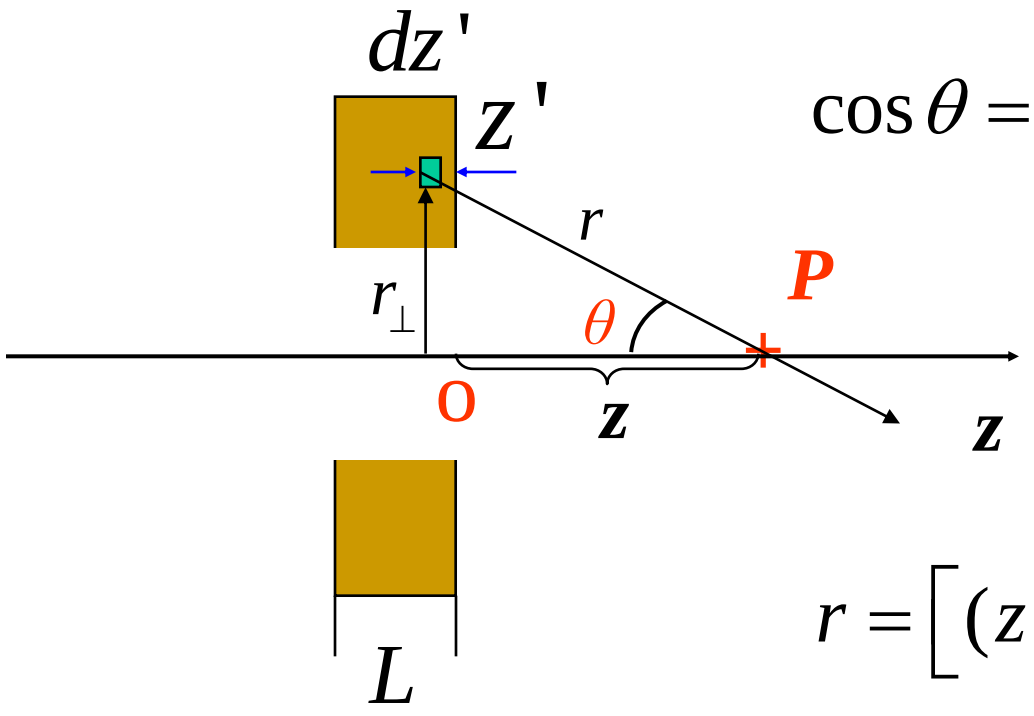
$$\boxed{\vec{E} = \int_{(q)} \frac{1}{4\pi\varepsilon_0} \frac{\rho dV}{r^2} \hat{r}}$$

例 均匀带电环柱轴线上的场强



$$\vec{E} = \vec{E}_z + \vec{E}_\perp$$

$$\boxed{\vec{E} = E_z \hat{z}}$$

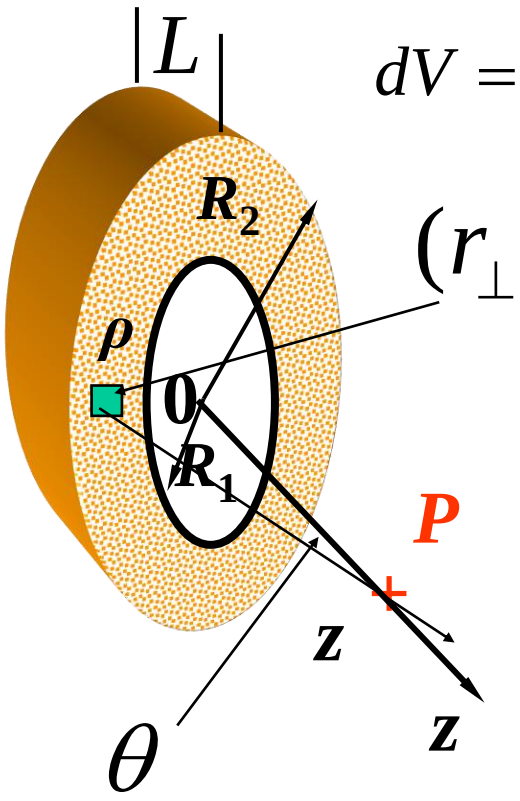


$$\cos\theta = \frac{z - z'}{r}$$

$$r = \left[(z - z')^2 + r_\perp^2\right]^{1/2}$$

$$dE_z = \frac{1}{4\pi\varepsilon_0}\frac{\rho dV}{r^2}\cos\theta$$

$$= \frac{1}{4\pi\varepsilon_0}\frac{(z - z')\rho dV}{\left[(z - z')^2 + r_\perp^2\right]^{3/2}}$$



$$dV = r_\perp d\varphi dr_\perp dz'$$

$$(r_\perp, \varphi, z')$$

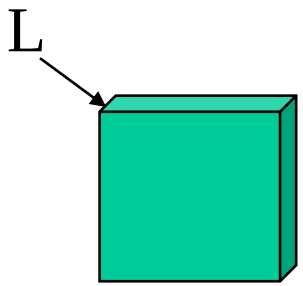
场强叠加原理

$$E_z = \iiint\limits_{\substack{r_\perp\{R_1 \to R_2\} \\ \varphi\{0 \to 2\pi\} \\ z'\{-L,0\}}} \frac{1}{4\pi\varepsilon_0} \frac{(z-z')\rho\, r_\perp dr_\perp d\varphi dz'}{\left[r_\perp^{\,2} + (z-z')^2\right]^{3/2}}$$

$$E_z = \frac{\rho}{2\varepsilon_o}[\sqrt{(z+L)^2 + R_1^2} - \sqrt{(z+L)^2 + R_2^2}$$

$$- \sqrt{z^2 + R_1^2} + \sqrt{z^2 + R_2^2}\,]$$

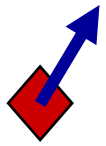
$$L \to 0$$



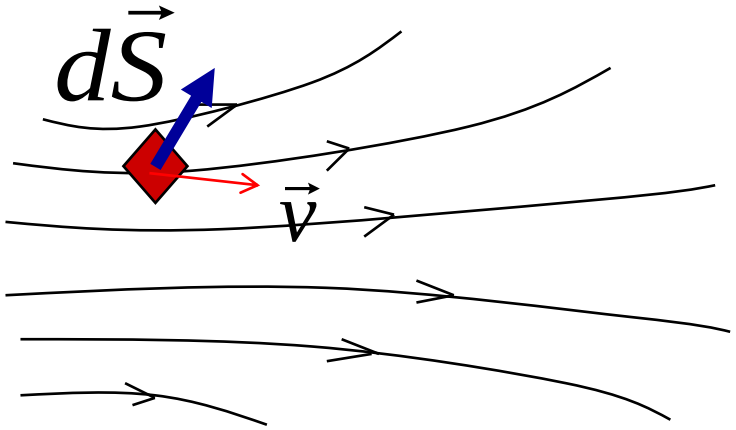
$$E_z = \frac{\rho L z}{2\varepsilon_o}[\frac{1}{\sqrt{z^2 + R_1^2}} - \frac{1}{\sqrt{z^2 + R_2^2}}]$$

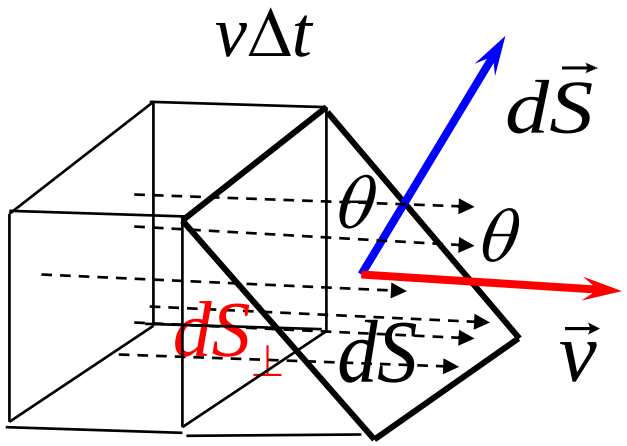
σ

1.5 电通量(electric flux)

面元  $d\vec{S} = \hat{n}dS$

类比流体场





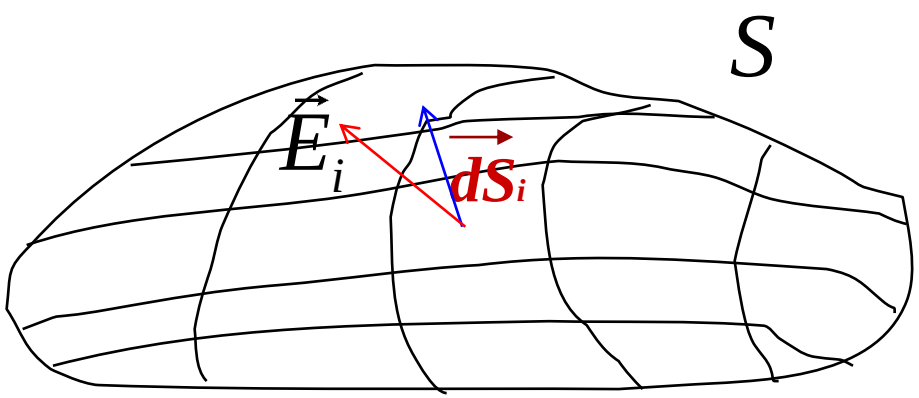
Δt 时间通过 $dS_\perp$ 的流体体积 $v\Delta t dS_\perp$

通量 $vdS_\perp = \vec{v}\cdot d\vec{S}$

电通量 $d\phi = \vec{E}\cdot d\vec{S}$

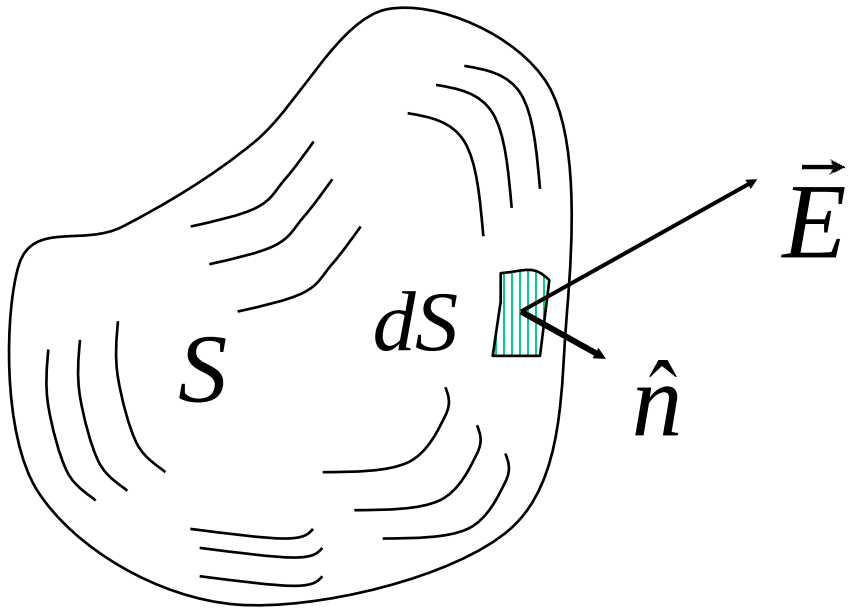
通过任意曲面的电通量

把曲面分成许多个面积元，每一面元处视为匀强电场



$$\phi = \sum_i \vec{E}_i \cdot d\vec{S}_i \equiv \iint_S \vec{E} \cdot d\vec{S}$$

❖ 通过闭合面的电通量

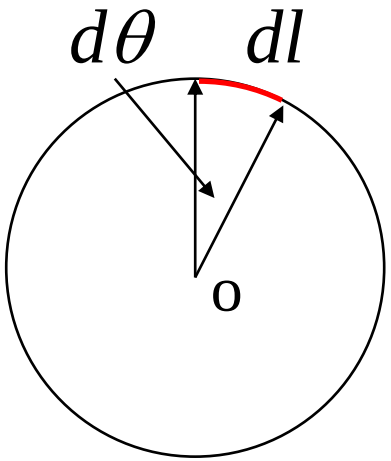


面元方向: 闭合面内指向面外

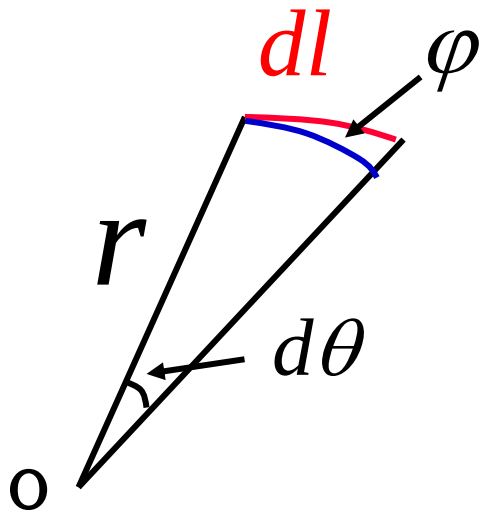
$$\phi = \sum_i \vec{E}_i \cdot d\vec{S}_i = \oiint_S \vec{E} \cdot \hat{n} dS$$

立体角*

平面角

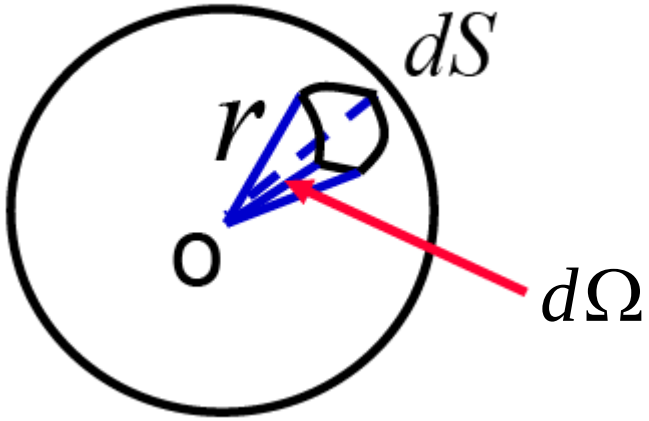


$$d\theta = \frac{dl}{r}$$

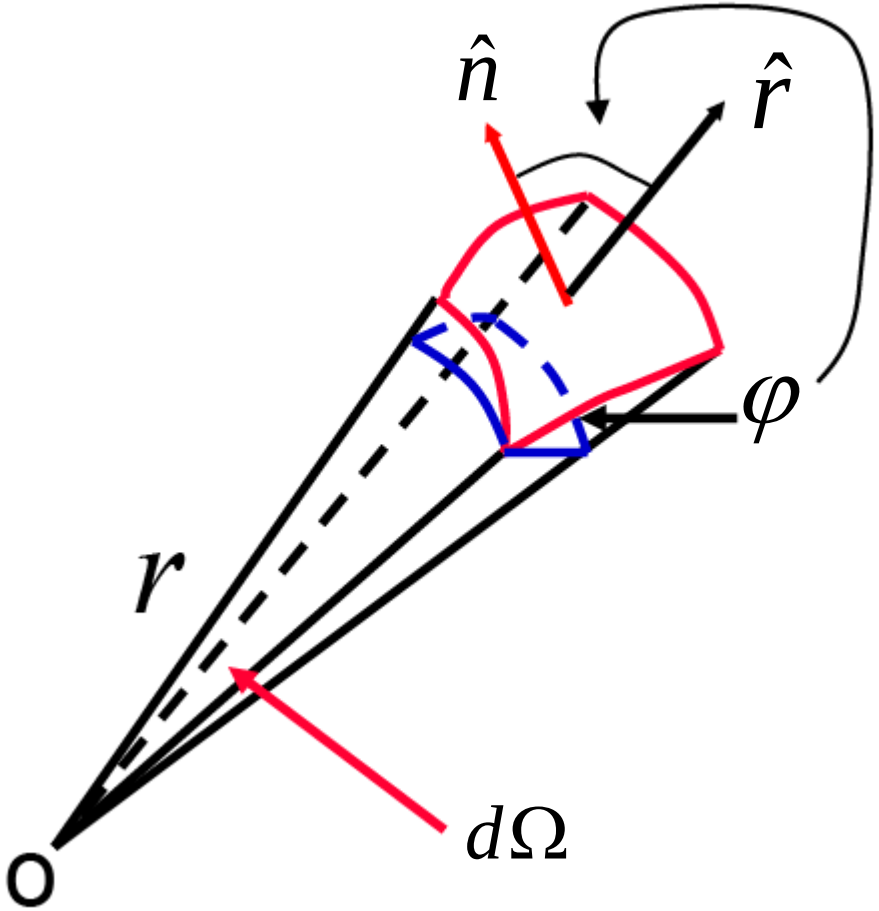


$$d\theta = \frac{dl\cos\varphi}{r}$$

立体角



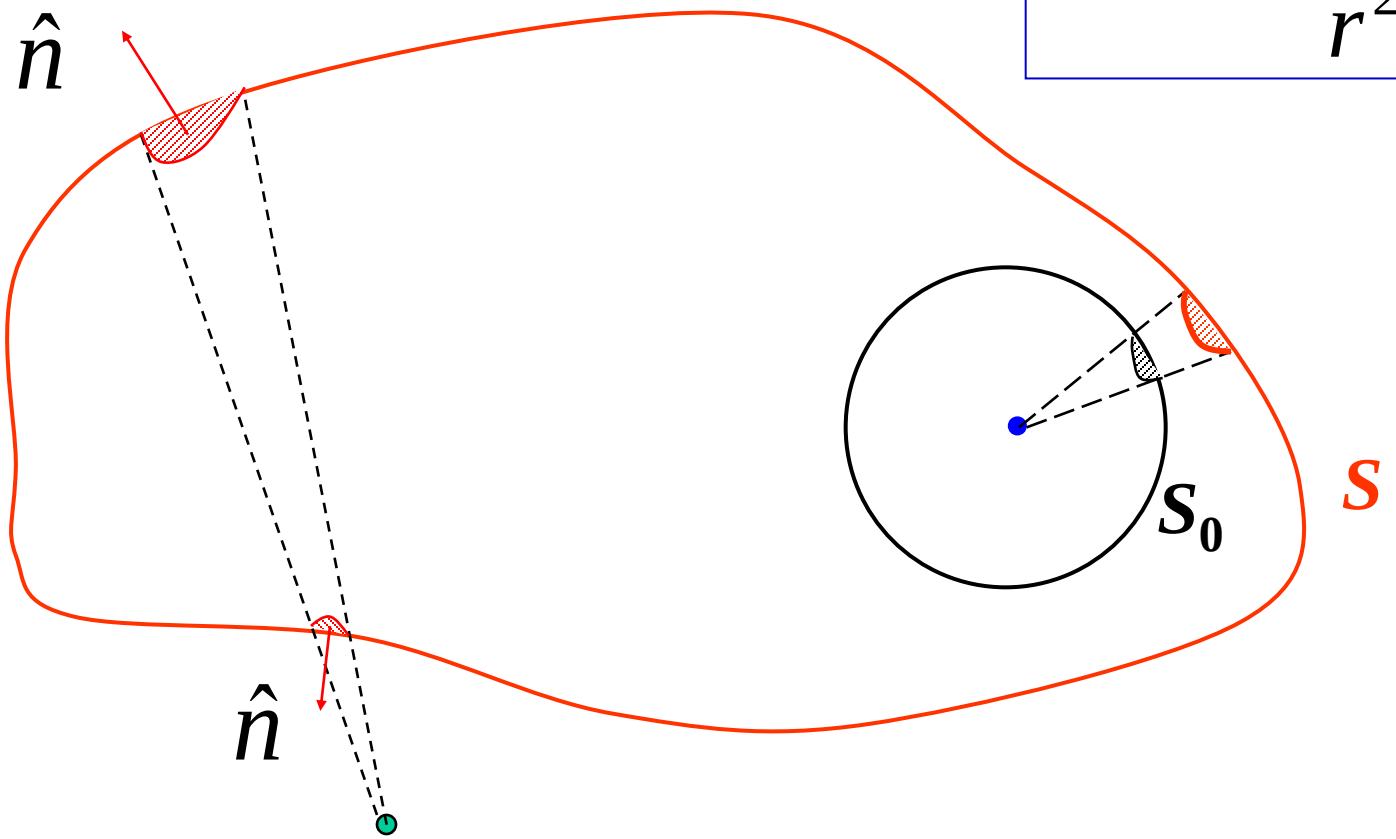
$$d\Omega = \frac{dS}{r^2}$$



$$d\Omega = \frac{dS\cos\varphi}{r^2} = \frac{\hat{r}\cdot\hat{n}dS}{r^2} = \frac{\hat{r}\cdot d\vec{S}}{r^2}$$

闭合曲面所张的立体角

$$d\Omega = \frac{\hat{r}\cdot d\vec{S}}{r^2}$$



闭合曲面外一点：$\Omega = 0$

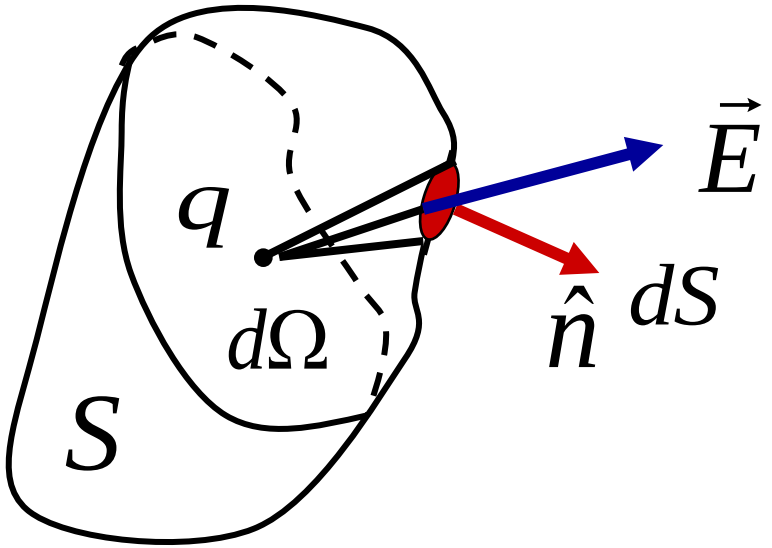
闭合曲面内一点：$\Omega = \oiint \frac{dS_0}{r_0^2} = 4\pi$

1.6 静电场的高斯定律证明*

在真空中的静电场内，任一闭合面的电通量等于这闭合面所包围的电量的代数和除以 ε_0

$$\boxed{\oiint_S \vec{E} \cdot d\vec{S} = \frac{\sum_i q_{i内}}{\varepsilon_0}}$$

闭合面包围一点电荷



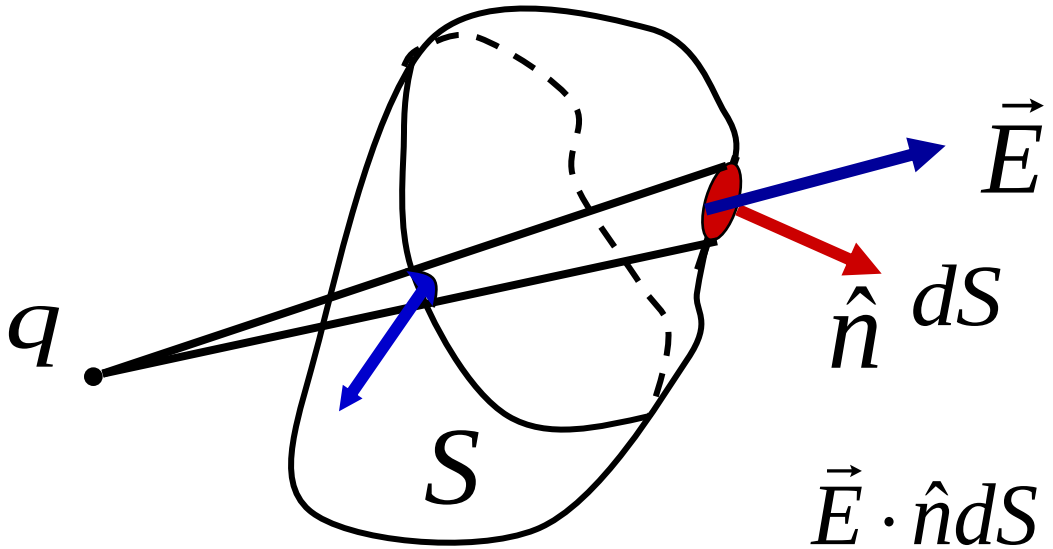
点电荷在面元处的场强为

$$\vec{E} = \frac{q}{4\pi\varepsilon_0 r^2}\hat{r}$$

$$\vec{E}\cdot\hat{n}dS = \frac{q}{4\pi\varepsilon_0 r^2}\hat{r}\cdot\hat{n}dS = \frac{q}{4\pi\varepsilon_0}d\Omega$$

$$\oiint_S \vec{E}\cdot\hat{n}dS = \frac{q}{4\pi\varepsilon_0}\oiint_S d\Omega = \frac{q}{\varepsilon_0}$$

点电荷在闭合面外



$$\vec{E}\cdot\hat{n}dS=\frac{q}{4\pi\varepsilon_0}d\Omega$$

$$\oiint_S \vec{E}\cdot\hat{n}dS=\frac{q}{4\pi\varepsilon_0}\oiint_S d\Omega=0$$

$$\oiint_S \vec{E}\cdot\hat{n}dS=\begin{cases}0 & \text{点电荷q在曲面外}\\ \dfrac{q}{\varepsilon_0} & \text{点电荷q在曲面内}\end{cases}$$

$$\oiint_S \vec{E}\cdot\hat{n}dS = \frac{q_{(S\text{曲面内电荷})}}{\varepsilon_0}$$

多个点电荷电场　　叠加原理

$$\oiint_S \vec{E}\cdot\hat{n}dS = \oiint_S \sum_i \vec{E}_i\cdot\hat{n}dS = \sum_i \oiint_S \vec{E}_i\cdot\hat{n}dS$$

$$\boxed{\oiint_S \vec{E}\cdot\hat{n}dS = \frac{\sum_i q_{内i}}{\varepsilon_0}} \xrightarrow{\text{连续分布}} \frac{\iiint_V \rho dV}{\varepsilon_0}$$

1.7 高斯定律和电场线

$$\boxed{\oiint_S \vec{E}\cdot\hat{n}dS = \frac{\sum_i q_{内i}}{\varepsilon_0}}$$

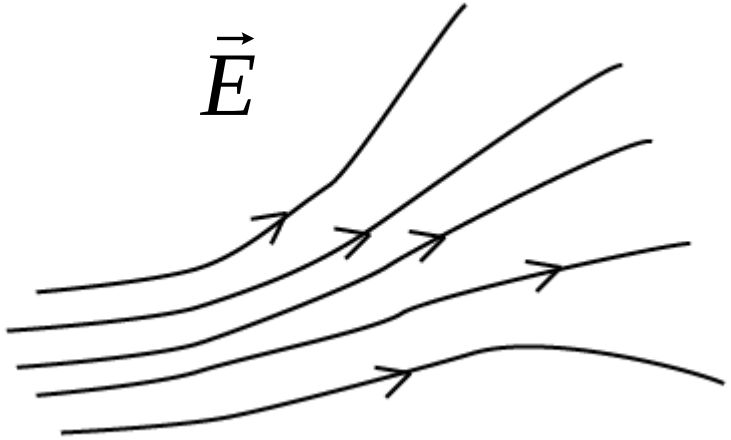
1. 闭合面内、外电荷对 $\vec{E}$ 都有贡献

2. 静电场性质的基本方程　有源场

3. 微分形式 $$\boxed{\nabla\cdot\vec{E} = \frac{1}{\varepsilon_0}\rho}$$

电场线(电力线)

用一族空间曲线形象描述场强分布, 把这些曲线称为电场线(electric field line)或电力线 (electric line of force)



1. 约定

方向：力线上每一点的切线方向

大小：密度正比于该点电场强度

电场线的性质

1) 电场线起始于正电荷(或无穷远处)，终止于负电荷，不会在没有电荷处中断；

(高斯定理)

2) 两条电场线不会相交；

(场的单值性)

3) 静电场的电力线不会形成闭合曲线。

(静电场环路定理)

【演示】

静电摆球

电场激发日光灯起辉

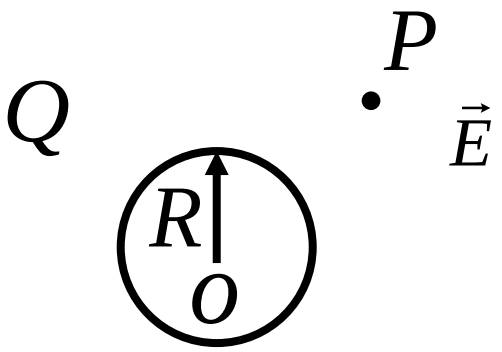
平面电荷的电场线

1.8 高斯定理的应用

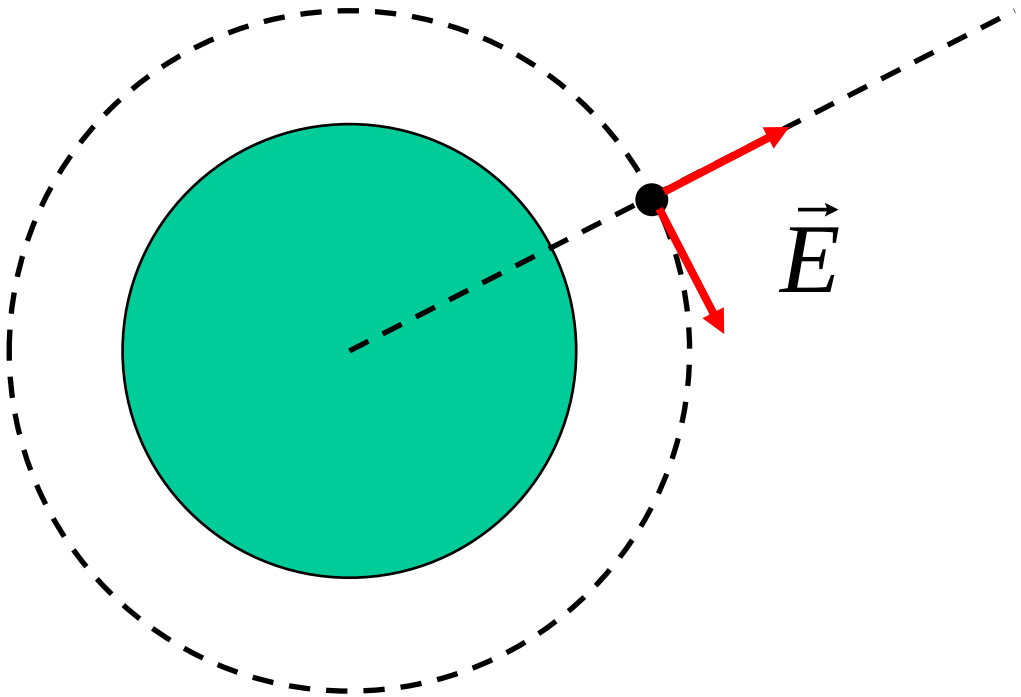
电荷的分布具有某种对称性的情况下利用高斯定理解电场比较方便

电场的对称性与电荷分布密切相关

例 均匀带电球面电场

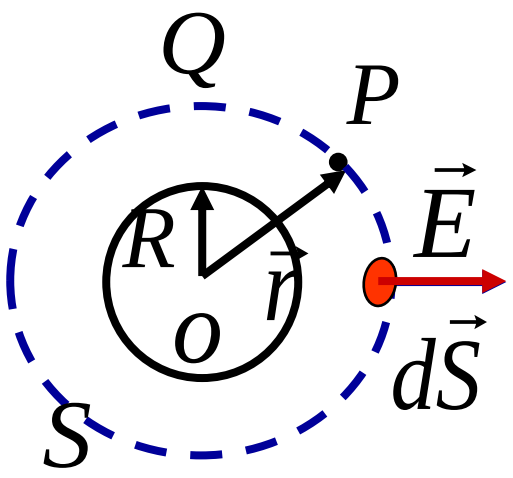


球对称



解:

球对称

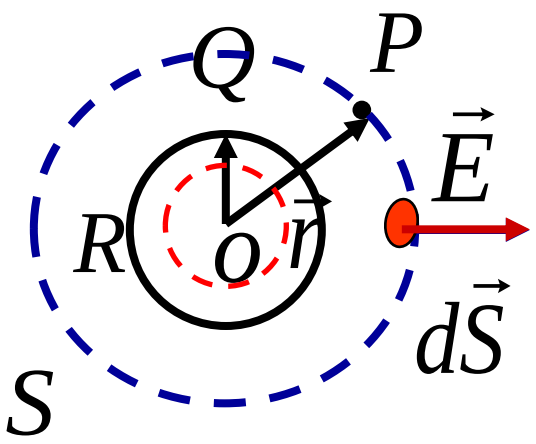


选取合适的高斯面(闭合面)

$$\oiint_S \vec{E}\cdot d\vec{S} = \oiint_S E dS = E\oiint_S dS = E4\pi r^2$$

$$= \frac{\sum_i q_i}{\varepsilon_0}$$

$$\boxed{E = \frac{\sum_i q_i}{4\pi\varepsilon_0 r^2}}$$



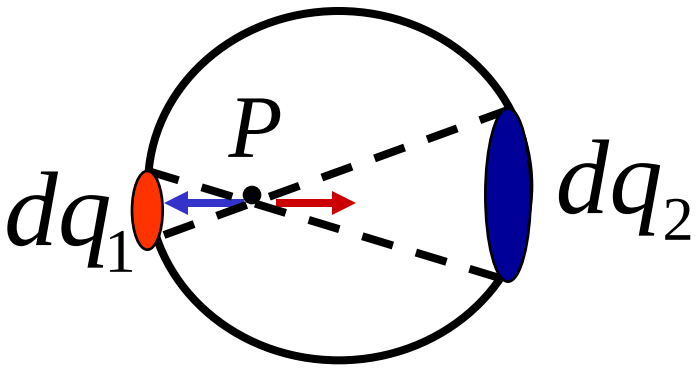
$$E=\frac{\sum_i q_i}{4\pi\varepsilon_0 r^2}=\begin{cases}0 & \text{球内}\\ \dfrac{Q}{4\pi\varepsilon_0 r^2} & \text{球外}\end{cases}$$

万有引力情况相似

如何理解面内场强为0 ？

过P点作圆锥

则在球面上截出两电荷元



$$dq_1 = \sigma dS_1 \qquad dq_2 = \sigma\, dS_2$$

$$dE_1 = \frac{\sigma dS_1}{4\pi\varepsilon_0 r_1^2} = \frac{\sigma}{4\pi\varepsilon_0} d\Omega$$

$$dE_2 = \frac{\sigma dS_2}{4\pi\varepsilon_0 r_2^2} = \frac{\sigma}{4\pi\varepsilon_0} d\Omega$$

相互抵消

均匀带电球面内电场为零与平方反比率密切相关

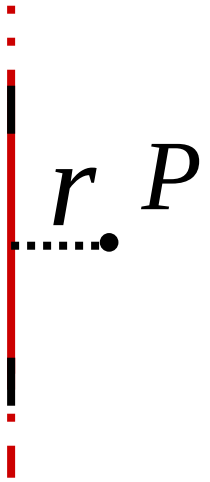
闭合导体壳内电场为零也是相关的一个结果

由此实验验证库仑定律

$$\propto \frac{1}{r^{2+\varepsilon}}$$

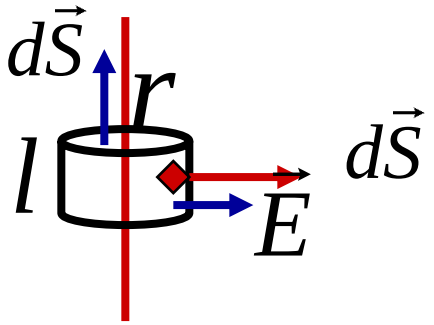
例 均匀带电的无限长的直线

线密度 λ



解：对称性的分析

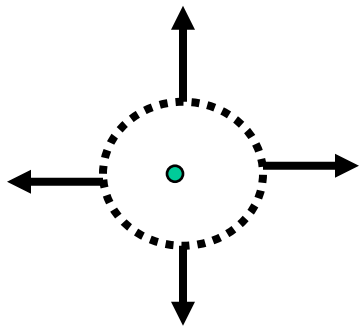
取合适的高斯面



$$\oiint_S \vec{E}\cdot d\vec{S} = \iint_{\text{侧面}} \vec{E}\cdot d\vec{S} + \iint_{\text{两底面}} \vec{E}\cdot d\vec{S}$$

$$= E2\pi rl$$

$$= \frac{\lambda l}{\varepsilon_0}$$



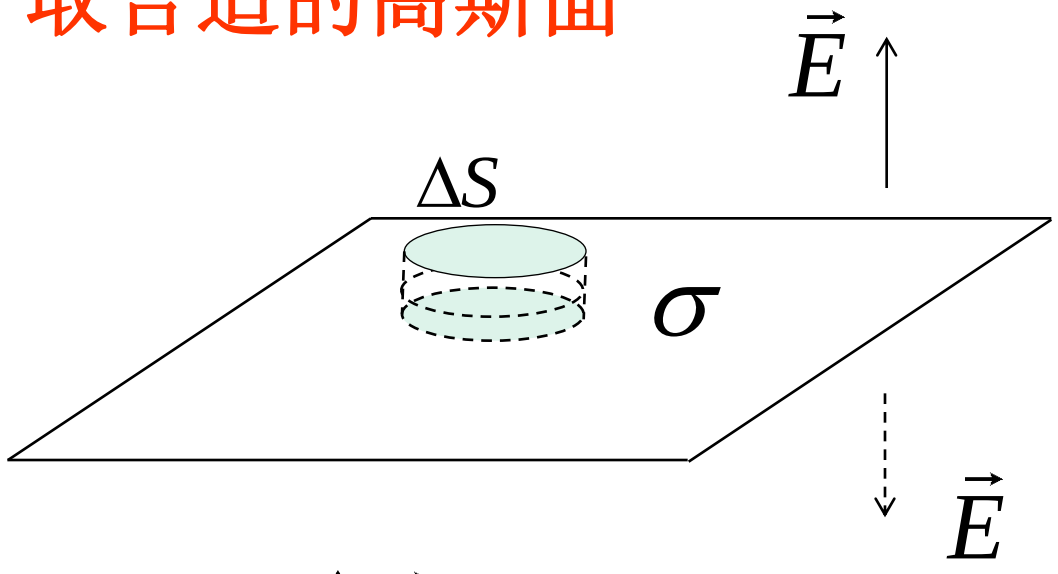
$$\boxed{E = \frac{\lambda}{2\pi\varepsilon_0 r}}$$

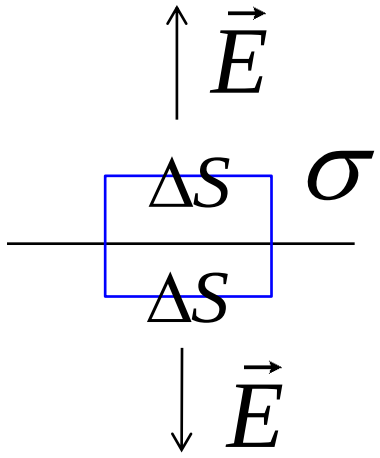
例 无限大均匀带电平板电场

面电荷密度 σ

解：对称性的分析

取合适的高斯面

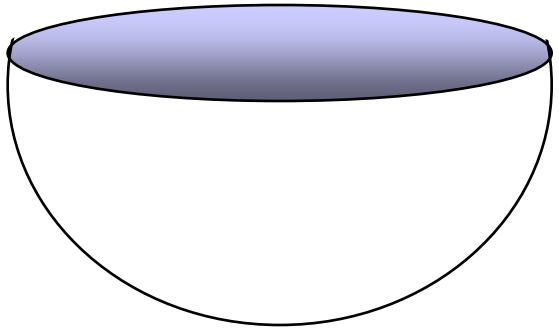




$$2E\Delta S = \frac{\sigma \Delta S}{\varepsilon_0}$$

$$E = \frac{\sigma}{2\varepsilon_0}$$

思考题



半球面上均匀带电，
大圆面上的电场方向？